ABCPROOF · MOTIZUKIPDF · FIELD GUIDE NO. 2

The Multiradial Algorithm

A field guide to [IUTchIII] Theorem 3.11 and Corollary 3.12: what goes in, what comes out, what gets blurred on the way, and why the blur turns into an inequality. With analogies, figures, real numbers, and an irresponsible amount of anime. （多輻的アルゴリズム完全攻略ガイド）

Running example: initial Θ-data for $1 + 8 = 9$ with $l = 7$ · three more worked examples · every number comes from the toolkit in `src/` · November 2026

ABSTRACT The multiradial algorithm is the engine of inter-universal Teichmüller theory. It takes a small piece of data that two incompatible "universes" of ring theory share, and from it rebuilds, in the language of *one* universe, the theta values that were built in the *other*, up to three controlled blurs called (Ind1), (Ind2) and (Ind3). Corollary 3.12 then measures the blurred picture with a ruler both universes agree on, the log-volume, and reads off an inequality. This guide walks through the algorithm step by step, in Mochizuki's own order (Theorem 3.11, the proof of Corollary 3.12, the steps (Stp1)–(Stp8) of [EssLgc] §3.10), and runs every step on four sets of initial Θ-data that the toolkit in this repository builds exactly. It collects the analogies Mochizuki himself uses (parallel transport, crystals, the Gaussian integral, a CPU and a network cable) and adds a few of our own, most of them from Japanese pop culture. It ends with an honest account of where the debate about Corollary 3.12 stands in November 2026, including the July 2026 interim report of the LANA formalization project.

CONTENT NOTICE

Jokes are text only: no character art, logos or screenshots. Where we say what the theory *claims*, we cite the papers in `pdf/`. Where people disagree about whether the claim is *proved*, we report each side in its own words and adjudicate nothing (§12). Numbers marked "toolkit" are produced by the C library in `src/` when this PDF is built; for curves this small they illustrate the shape of the computation, not a sharp bound (§9.4). 審判ではありません。

Contents

1 The plot in one breath
2 Why this is hard: the Θ-link breaks the ring
3 Multiradiality: the one definition you need
4 The stage: the log-theta-lattice
5 The cast: every object the algorithm touches
6 The algorithm, step by step (Theorem 3.11)
7 The three blurs: (Ind1), (Ind2), (Ind3)
8 From picture to inequality (Corollary 3.12)
9 The numbers: four worked examples
10 Logic: why "AND" and not "OR"
11 Analogy gallery
12 Where the debate stands (November 2026)
13 Run it yourself
14 Glossary and references

HOW TO READ THIS GUIDE

Three speeds. **Express** (ten minutes): read §1, look at Figure 6.1 and Figure 8.1, read the boxed "key" statements. **Local** (an evening): read §1–§9 in order and skip the green paper boxes. **Kakueki-teisha 各駅停車** (stops at every station): read everything and keep [IUTchIII] open at pages 153–186. Boxes are colour-coded: pink for jokes, purple for analogies, green for what the papers say, blue for toolkit output and yellow for caveats. Nothing mathematical rests on a joke, and no joke rests on anything mathematical.

1 The plot in one breath 一息であらすじ

Rebuild the theta values of one universe out of the parts of it that survive the trip into another universe, accept a little blur, and then weigh the result with a scale that both universes trust.

That sentence is the multiradial algorithm and its corollary. The rest of this guide unpacks it. First, why anyone would want to do this.

1.1 The N-th power trick

Take a nonzero integer x and measure its size by its height $h(x) = \log|x|$. Raising to the N-th power multiplies the height by N:

$$h(x^N) \;=\; N \cdot h(x).$$

Now suppose someone hands you a machine that, given only some data that x and x^N have *in common*, rebuilds the size of x^N in the coordinates of x, and is wrong by at most ε. Then $N\,h(x) \le h(x) + \varepsilon$, so

$$h(x) \;\le\; \frac{\varepsilon}{N-1}.$$

A machine that "cannot tell x from x^N up to a small error" bounds the height. This is the whole strategy of inter-universal Teichmüller theory in one line, and it is how Mochizuki himself summarizes it in his April 2026 progress report on formalizing the theory in Lean [Form26, §3–§4]: the essential content of Corollary 3.12 is "that the height of the elliptic curve under consideration is equal to N times the height of the elliptic curve (where N is a large positive number), up to a small discrepancy (arising from (Ind1,2,3) + hull), thus implying a bound on the height."

Of course no such machine exists for integers inside ordinary arithmetic: $x \mapsto x^N$ is not a ring homomorphism of $\mathbb{Z}$, and nothing about $\mathbb{Z}$ "forgets" the difference between x and x^N. The theory's answer is to put x and x^N in *two different copies* of arithmetic, glue the copies along the part they can share, and then build the machine. The gluing is the Θ-link. The machine is the multiradial algorithm.

1.2 What plays the role of x and x^N

In IUT the role of x is played by the **q-pilot**: the $2l$-th roots $\underline{q}_v$ of the Tate parameters of the elliptic curve at its bad places v. The role of x^N is played by the **Θ-pilot**: the collection of theta values

$$\left\{ \underline{q}_v^{\,j^2} \right\}_{j=1,\ldots,l^*}, \qquad l^* = \tfrac{l-1}{2},$$

one for each nonzero label j up to sign. So there is not one exponent N but a whole row of them, $1, 4, 9, \ldots, (l^*)^2$, and their average is

$$\frac{1}{l^*}\sum_{j=1}^{l^*} j^2 \;=\; \frac{l(l+1)}{12} \;=\; 4.6667 \quad (l=7).$$

Both pilots have a log-volume, which is (minus) their arithmetic degree. For our running example $1 + 8 = 9$ the q-pilot has log-volume $-|\log \underline{q}| = -0.3139$ and the Θ-pilot has naive log-volume $-1.4648 = -4.6667 \cdot |\log \underline{q}|$. The Θ-link declares these two objects to be "the same" (as objects of an

abstract monoid, not of a ring), and the multiradial algorithm then shows that the declaration costs almost nothing.

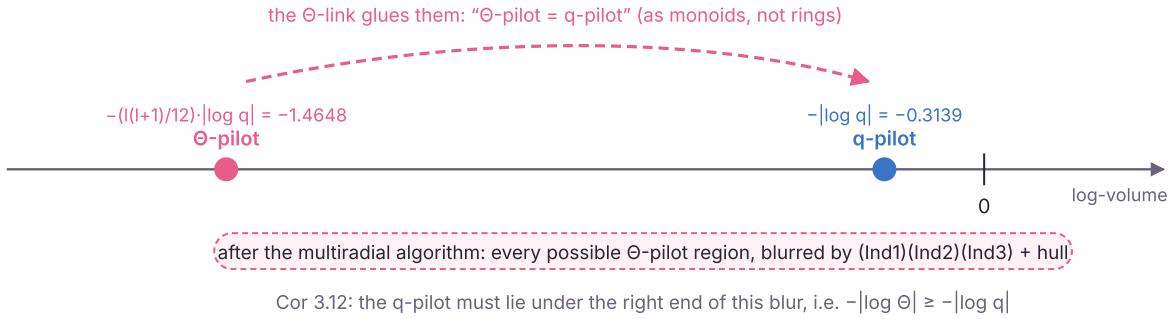


Figure 1.1. The plot on one number line (not to scale). The Θ-link pretends a big object (the Θ-pilot, "x^N") and a small one (the q-pilot, "x") are the same. The multiradial algorithm shows that this pretence is consistent once you blur the Θ-pilot a little. Corollary 3.12 says the blur cannot be small and still miss the q-pilot, which is an upper bound on how far apart they were.

1.3 The three acts

act	what happens	where	difficulty
1. The Θ-link	Glue two copies of arithmetic (two Θ±ell NF-Hodge theaters) along a shared prime-strip, sending the Θ-pilot of one to the q-pilot of the other.	[IUTchI] Cor 3.7, [IUTchII] Cor 4.10, [IUTchIII] Def 3.8	a definition
2. The multiradial representation	Rebuild the Θ-pilot from the shared data alone, in a form readable from both sides, up to (Ind1), (Ind2), (Ind3).	[IUTchIII] Thm 3.11	the bulk of [IUTchI–III]
3. The log-volume estimate	Wrap the blurred picture in a holomorphic hull, measure it, and compare with the q-pilot: $-\lvert\log\Theta\rvert \geq -\lvert\log q\rvert$.	[IUTchIII] Cor 3.12	"a few lines" per step, says [EssLgc] §3.11
4. (Encore) the numbers	Compute the left side explicitly; get a Szpiro-type height bound.	[IUTchIV] Thm 1.10, Cor 2.2	long but elementary

Mochizuki's own summary of the logical flow is exactly this three-act structure [EssLgc, Ex. 2.4.5 (i)]: one starts with the definition of the Θ-link; one then constructs "a complicated apparatus that is referred to as the multiradial representation of the Θ-pilot"; finally one derives a numerical estimate "in an essentially straightforward fashion". Section 10 below explains his toy model of this flow with three real numbers.

OTAKU CORNER · 君の名は。

The cleanest anime analogy for the Θ-link is the body-swap film *Your Name* (Kimi no Na wa, 2016). Two teenagers live in two towns, three years apart, and wake up in each other's bodies. What they share is the body (the prime-strip); what they do not share is the town, the year, the friends, the whole "ring structure" of daily life. Neither can phone the other, because their worlds do not even agree on the date. So they leave notes on the phone, written so that they make sense *whoever* reads them. Those notes are the multiradial representation: messages about one world, written in terms that are valid in both ("simultaneous holomorphic expressibility", §6.6). And the film's catastrophe is only avoided because of what both sides can agree on: a comet with a measurable size, much like the log-volume that Corollary 3.12 reads off at the end.

WHAT THE PAPER SAYS · [IUTCHIII] REMARK 3.11.1 (I)

"Theorem 3.11 gives an algorithm for describing, up to certain relatively mild indeterminacies, the LGP-monoids — i.e., in essence, the theta values $\{\underline{q}^{j^2}\}_{j=1,\ldots,l^*}$ — which are constructed relative to the scheme/ring structure, i.e., 'arithmetic holomorphic structure', associated to one vertical line [...] in the LGP-Gaussian log-theta-lattice under consideration, in terms of the a priori alien arithmetic holomorphic structure of another vertical line."

2 Why this is hard: the Θ-link breaks the ring 環が壊れる理由

A ring has two operations, $+$ and $\times$, welded together by the distributive law. [Alien] §2.11 calls them the "two underlying combinatorial dimensions of a ring", and the image to keep in mind is a sheet of paper: multiplication runs one way, addition the other, and the distributive law is what keeps the sheet from tearing. For a local field k the multiplicative direction itself splits again into a *unit* part $\mathcal{O}_k^\times$ (things of size 1) and a *value group* part (the powers of a uniformizer, which carry the size).

The map we want, $\underline{q}^{j^2} \mapsto \underline{q}$, only makes sense on the value-group part. It is a perfectly good map of multiplicative monoids, but there is no ring homomorphism behind it: in characteristic zero $(a+b)^N \neq a^N + b^N$, so any map that dilates sizes by N tears the sheet. That is why the Θ-link cannot be a map of rings, of schemes, or of anything that remembers $+$. It is a gluing of two separate copies of arithmetic along the part that has no $+$ in it.

ANALOGY · CHARACTERISTIC P HAS IT EASY

In characteristic p the map $x \mapsto x^p$ *is* a ring homomorphism (Frobenius), because $(a+b)^p = a^p + b^p$ there. So over function fields the N-th power trick of §1.1 can be run inside one copy of arithmetic, and the geometric Szpiro inequality has proofs along exactly these lines (via Frobenius, or via the Kodaira–Spencer map) [Alien, §2.4–§2.9]. Number fields have no Frobenius. IUT's move is to build a substitute for Frobenius that is allowed to leave the universe: the Θ-link. Mochizuki's 2026 report gives the simplest model: "when $N = p$, working modulo p [...] there is a common ring structure that is compatible with the p-th power map" [Form26, §3]. Over $\mathbb{Z}$ there is no such common ring structure, so the theory has to manufacture a common *container* instead.

2.1 What goes through the Θ-link and what does not

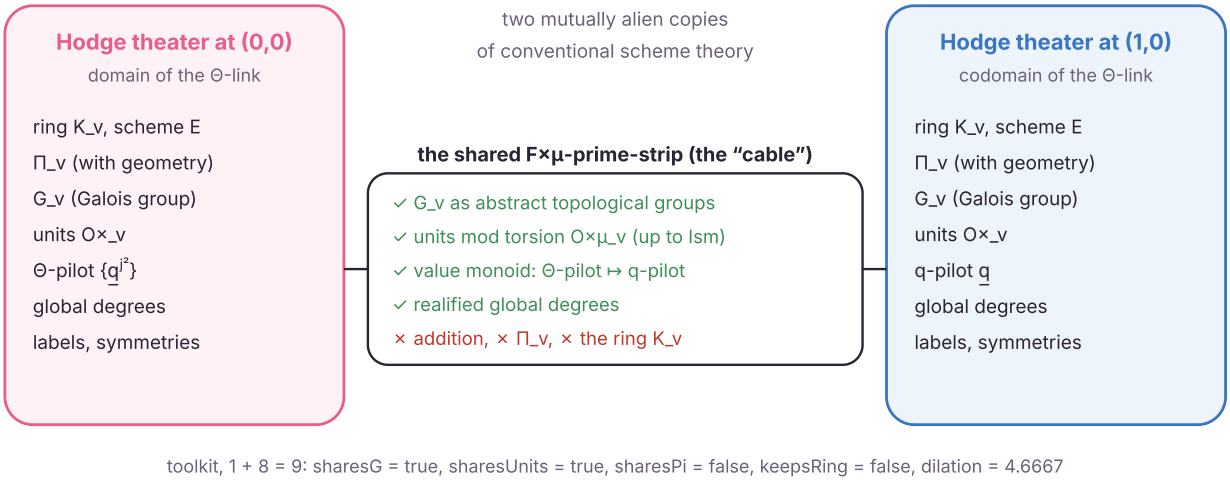


Figure 2.1. The Θ×μ_LGP-link of [IUTchIII] Definition 3.8 (ii) as the toolkit's lattice records it. Only the F×μ-prime-strip passes from one side to the other, by a *full poly-isomorphism* (every isomorphism at once, no preferred one). What it carries: the Galois groups G_v as abstract groups, the units modulo torsion with their integral structures, and the value-group monoid, along which the Θ-pilot on the left is identified with the q-pilot on the right. What it does not carry: addition.

So after the Θ-link we have a strange situation. On the right there is an honest elliptic curve over an honest number field, with its q-pilot. On the left there is another honest elliptic curve over another honest number field, with its Θ-pilot. And there is a bridge between them that carries only abstract groups and multiplicative monoids. The question of Theorem 3.11 is:

> **THE QUESTION THEOREM 3.11 ANSWERS**
> Can the person standing on the **right** describe the Θ-pilot of the **left** in their own ring theory, using only what came across the bridge? If so, how precisely?

Answer: yes, up to three explicit blurs, and the description is good enough to measure. The tools that make this possible are exactly the ones developed in the decade before IUT: *anabelian geometry* (rebuild a ring-like object from its Galois group alone), *Kummer theory* (relate an object to its portable photograph in Galois cohomology), the *étale theta function* (whose Kummer theory has an unusually rigid cyclotome), and the *p-adic logarithm* (which turns the multiplicative units into an additive module, and so manufactures a place where addition can live again). The companion guide *anabelian_guide.pdf* in this repository covers the first three in depth.

> **OTAKU CORNER · 異世界転生**
> Every isekai protagonist faces the problem of Theorem 3.11. You are hit by a truck (Truck-kun, the genre's most prolific transport mechanism) and wake up in a world whose laws of nature differ from yours. All you carried across is what fits in your pockets: in IUT, an abstract group G_v and a bag of units. The ring structure, your house, the power grid, stayed behind. The plot of a hundred light novels is then: rebuild modern civilization from what you carried, in a way that also works in the new world. IUT's version is stricter than any light novel: what you rebuild must be described so that a native of *either* world would accept it, and you have to say exactly how wrong it might be. No cheat skill (チート能力) is granted, although the étale theta function comes close.

2.2 Why not just compare the two sides directly?

Because there is nothing to compare them in. To say "the Θ-pilot on the left has log-volume -1.4648 and the q-pilot on the right has log-volume -0.3139" one needs a common copy of the real numbers into which both log-volumes map, and a common ring structure in which both pilots are honest arithmetic line bundles. The Θ-link provides neither: it identifies the two pilots only as abstract objects of a monoid, up to isomorphism class. Any argument of the form "compute both sides and compare" silently assumes a common ring structure, which is precisely what the Θ-link destroyed.

Mochizuki's 2026 report puts it as a diagram [Form26, §3]: for two ring structures ${}^{\dagger}R$ and ${}^{\ddagger}R$ glued along their multiplicative monoids, the multiradial representation is a common container

$$ {}^{\dagger}R \;\subseteq\; \big(\text{multiradial representation of the } \Theta\text{-pilot}\big) \;\supseteq\; {}^{\ddagger}R, $$

built from "the multiplicative monoid and abstract group structures (but not the ring structures!)" that are "common (cf. 'logical AND $\wedge$!') to $\dagger$, $\ddagger$". The rest of this guide is about how that container is built and what it costs.

3 Multiradiality: the one definition you need 多輻性とは

The word "multiradial" comes from a picture of a wheel. The hub is something shared (the *coric* data, from "core"), and the spokes are the different richer structures that sit over it (the *radial* data). The question is whether the hub can be turned without breaking the spokes.

DEFINITION · [IUTCHII] EXAMPLE 1.7; [ALIEN] §3.1 (II)–(III)
A **radial environment** is a triple: a type of **radial data** $\mathcal{R}$, a type of **coric data** $\mathcal{C}$, and a functorial **radial algorithm** $\mathcal{R} \to \mathcal{C}$ that forgets the finer structure. It is **multiradial** if the resulting functor is *full*: every isomorphism between the coric data underlying two radial data lifts to an isomorphism of the radial data. It is **uniradial** otherwise.

Fullness has a vivid consequence, the **switching property** [Alien, §3.1 (iii)]: take two radial data R_1, R_2 together with an isomorphism α between their coric data. Swapping the roles of R_1 and R_2 (and replacing α by α^{-1}) gives an isomorphic configuration. Multiradiality is equivalent to every such pair being isomorphic to a "diagonal" pair (R, R, id). In words: *once you only look through the hub, you cannot tell which spoke is which.*

3.1 The prototype: complex structures on a real plane



Figure 3.1. [Alien] Example 3.2.1, after its Figures 3.2 and 3.3. A complex line, viewed as a real plane, is uniradial: a random real-linear automorphism of the plane does not respect multiplication by i. If instead the radial data is the plane together with the whole $GL_2(\mathbb{R})$-orbit of complex structures, every automorphism lifts. The price is that you no longer know *which* complex structure you have; the space of possibilities is the upper and lower half-planes, i.e. classical Teichmüller space. This is where the "Teichmüller" in IUT comes from.

Note what just happened. Making something multiradial is *easy* if you are willing to throw information away: replace your structure by the set of all structures it could be. The tautological recipe even has a name, **multiradialization** [IUTchII, Ex. 1.7 (ii)]. The art of Theorem 3.11 is to throw away exactly enough. The blurred description must still determine the one number we care about, the log-volume, up to an error we can bound. Everything in §6 and §7 is about that balance.

> **THE WHOLE GAME IN ONE SENTENCE**
> Find a description of the Θ-pilot that is (1) **multiradial**, so that the person on the other side of the Θ-link can read it, and (2) still **informative about log-volumes**, so that reading it tells them something. Either one alone is trivial. Both together is Theorem 3.11.

3.2 Multiradiality as parallel transport

Mochizuki's favourite way to think about multiradiality is through connections [IUTchII, Rem. 1.7.1; Alien, §3.1 (iv)]. Grothendieck's definition of a connection on a bundle E over X/S is an isomorphism between the fibres of E at two infinitesimally close points of X lying over the same point of S. Read this as: two "radial" fibres over the same "coric" base point, and a rule for transporting one to the other. Fullness of the radial functor says such transports exist along every path in the coric data; the leftover choice in the lift is a kind of **monodromy**. In IUT, that monodromy is precisely the indeterminacies (Ind1), (Ind2), (Ind3).

The analogy goes further. [Alien] §3.1 (v) rewrites the classical proof of the geometric Szpiro inequality via the Kodaira–Spencer map in these terms: a rank-two bundle $\mathcal{E}$ with a logarithmic connection serves as a *container* for the line bundle $\omega_E \subseteq \mathcal{E}$. Transport ω_E from one factor of $S \times S$ to the other through the container and measure how far it moves; the discrepancy *is* the Kodaira–Spencer map, and its non-vanishing bounds $\deg \omega_E$. The multiradial representation is the arithmetic version: tensor packets of log-shells are the container, the Θ-pilot is the line bundle, and the blur is the discrepancy.

ANALOGY · KARAOKE KEY CHANGE

The backing track is coric: everyone in the karaoke box hears the same one. The singer's voice is radial. A song is “uniradial” if it only works in the original key. It is “multiradial” if every transposition of the backing track (the machine's ♯ / ♭ buttons, which act on the coric data) can be matched by the singer. The tautological way to be multiradial is to hum tunelessly, which matches every key and says nothing. The Theorem 3.11 way is to sing so that the *loudness* of the performance (the log-volume) is the same in every key, even if you cannot say which key you are in.

3.3 A catalogue of radial environments in IUT

The toolkit's `ClassifyRadial` process lists the radial environments that appear in the papers, with a verdict for each. Read it top to bottom as the story of the theory: the ring is uniradial, the Galois groups are tautologically multiradial, the units modulo torsion become multiradial if you accept (Ind2), the full Kummer structure on the units is uniradial (it needs a cyclotome), the mono-theta environment is multiradial (its cyclotome is rigid), and the full multiradial representation of Theorem 3.11 is multiradial up to (Ind1)–(Ind3).

TOOLKIT · IUT RUN CLASSIFYRADIAL

```
ClassifyRadial  [IUTchII] Ex 1.7-1.9 — radial environments, multiradiality, multiradialization
  in: radial data, coric data   out: multiradial / uniradial
    uniradial    R = holomorphic structure: Π_v ↷ O▷_{F̃_v} (an F-prime-strip) | C = D⊢ = {G_v}   [IUTchII] Ex 1.8 (i):
uniradial — the ring structure is not transportable
    multiradial  R = D⊢ itself | C = D⊢   [IUTchII] Ex 1.8 (ii): the coric data is tautologically multiradial
    multiradial  R = O×μ_v with G_v-action, up to Ism | C = D⊢   [IUTchII] Ex 1.8 (iv), Cor 1.12: units modulo torsion
are multiradial (Ind2)
    uniradial    R = O×_v with its Ẑ×-Kummer structure | C = D⊢   [IUTchII] Ex 1.8 (iii): the full Kummer structure on
O× is uniradial (needs the cyclotome)
    multiradial  R = the mono-theta environment and its cyclotomic rigidity (l·Δ_θ)(Π) ≅ Π_μ | C = D⊢   [IUTchII] Cor
1.10, Ex 1.9: multiradial
    multiradial  R = theta monoids Ψ_env(D_>) with their splittings | C = (D⊢, F⊢×μ(D⊢))   [IUTchII] Thm 2.2, Cor 2.3
(étale-picture radial environment)
    multiradial  R = the multiradial representation R^LGP: packets, splitting monoids, (F_mod^×)_j | C = (D⊢, F⊢×μ)
[IUTchIII] Thm 3.11 (i): multiradial up to Ind1, Ind2, Ind3
    multiradialization (Ex 1.7 (ii)): uni → multi (α = full poly-iso of D⊢ adjoined)
```

OTAKU CORNER · 推し

Fandom has its own radial environment. The coric data is the anime as broadcast; the radial data is each fan's interpretation. An interpretation is “multiradial” if it survives every re-edit of the broadcast: director's cut, Blu-ray fixes to the animation, the movie recap. Most fan theories are uniradial and die with the Blu-ray release. The rare multiradial ones are vague enough to survive and specific enough to predict the box-office numbers. Theorem 3.11 is the fan theory that predicts the box office.

4 The stage: the log-theta-lattice 舞台装置

The algorithm runs on a two-dimensional grid of complete copies of arithmetic, the **log-theta-lattice** [IUTchIII, Def. 1.4, Def. 3.8 (iii)]. Each node (n, m) is a Θ±ell NF-Hodge theater: one whole copy of “conventional scheme theory” around the elliptic curve, with its number fields, local fields, Galois groups, Frobenioids, and the label symmetries $\mathbb{F}_l^{\times\pm}$ and $\mathbb{F}_l^{*}$ (see the Hodge theaters infographic in `docs/`). Different nodes live in different universes, so nothing in one node is literally an element of another.

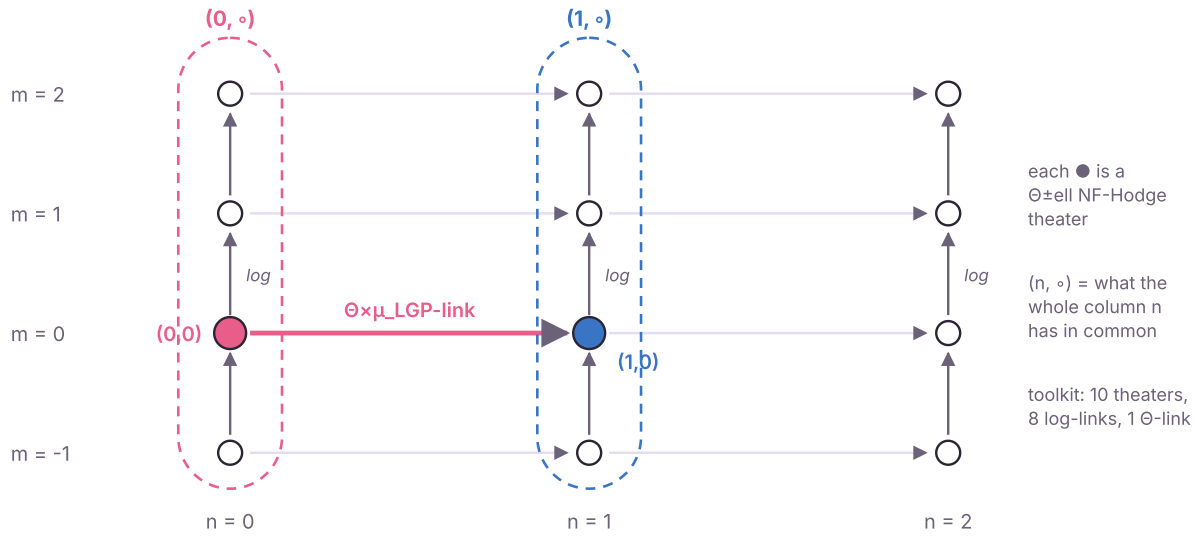


Figure 4.1. The part of the LGP-Gaussian log-theta-lattice the algorithm uses: two columns of vertical log-links and one horizontal Θ×μ_LGP-link from $(0,0)$ to $(1,0)$. Mochizuki calls this shape the "infinite H" [EssLgc, §3.3 (InfH)]. A dashed capsule $(n,\circ)$ stands for what all the Hodge theaters of column n have in common: the vertically coric data.

4.1 Two kinds of arrows

arrow	what it does	what it shares (coric)	what it scrambles
log-link ↑	At each place v, the p-adic (or complex) logarithm sends the units of the node below to an additive module that becomes the multiplicative world of the node above.	the arithmetic fundamental groups Π_v and G_v (vertical coricity, [IUTchIII] Thm 1.5 (i))	which elements are units and which are values: it "juggles" ⊞ and ⊠
Θ-link →	Glues the Θ-pilot of (n,m) to the q-pilot of $(n+1,m)$ as abstract monoids.	G_v and the units modulo torsion $\mathcal{O}^{\times\mu}$ (horizontal coricity, Thm 1.5 (ii))	the ring structure and Π_v: the two sides are alien

Two facts about the arrows drive the whole algorithm.

The lattice does not commute. Going up a log-link and then across a Θ-link is not the same as going across and then up. The Lean port in `lean/IUT/Link.lean` proves this for the toolkit's data model as the theorem `square_not_commutative`. Consequence: one must use a *single* Θ-link and do everything else vertically. Using the unit part of one Θ-link and the value part of another would require a commuting square that does not exist [IUTchIII, proof of Cor. 3.12, Step (iii)].

Some data survive both kinds of arrow. The Galois groups G_v, and the units modulo torsion $\mathcal{O}^{\times\mu}$ rebuilt from them, are shared vertically *and* horizontally: they are **bi-coric** [IUTchIII, Thm 1.5 (iii)]. So are the **log-shells** built from them (Thm 1.5 (iv)). These are the only objects that can serve as a common container for both columns, and Theorem 3.11 uses them exactly that way.

TOOLKIT · IUT RUN CHECKBICORIC

```
CheckBicoric  [IUTchIII] Thm 1.5 — vertical / horizontal / bi-coric data of the lattice
  in: LogThetaLattice   out: bicoric summary
    (i) D-HT vertically coric 1 (Π copy 170002)  (ii) F⊢×μ horizontally coric 1 (units copy 183003)  (iii) F⊢×μ(D⊩_Δ)
bi-coric 1  (iv) log-shells bi-coric 1  (v) D⊪(D⊩) bi-coric 1
```

4.2 Log-shells: the container that holds the log of everything

For a nonarchimedean place v over an odd prime p, the log-shell is

$$\mathcal{I}_v \;=\; \tfrac{1}{2p}\,\log_p\!\big(\mathcal{O}_{K_v}^{\times}\big) \;\subseteq\; K_v,$$

a compact additive lattice that contains the logarithms of all units, with a little room to spare ([IUTchIII] Def. 1.1, Prop. 1.2). Its purpose is to be an *upper bound* that is stable under the log-link: whatever level of the column you are on, the logarithm of the units at the next level lands inside the log-shell at this one. That stability is what makes log-shells "mono-analytic containers" [IUTchIII, Rem. 1.5.2], and it is also the reason one of the three blurs, (Ind3), is only an inclusion "$\subseteq$" rather than an equality (§7.3). At archimedean places the log-shell is a closed disc and the inclusion becomes a surjection "$\twoheadrightarrow$".

TOOLKIT CAVEAT

The initial Θ-data JSON files carry no ramification data for the field K, so the toolkit takes $e = 1$ everywhere and every log-shell is $p^0 \cdot \mathcal{O}$ with log-volume 0. The shape of the algorithm is unaffected; the sizes of the containers come in only through the estimate of $\log d^K$ (§9.4).

OTAKU CORNER · 大道芸

Mochizuki's word for what the log-link does to addition and multiplication is *juggling* [IUTchIII, Rem. 1.2.2 (vi)]. Picture a daidōgei street performer in Asakusa: two clubs, ⊞ and ⊠, constantly swapping hands. Each throw (log-link) turns the multiplicative club into the additive one. The audience (the coric data) only ever sees the performer's hands, which do not change; the clubs are never in the same hand twice. The log-shell is the performer's reach: whatever is in the air, it is within that circle. The non-commutativity of the lattice is the fact that you cannot juggle and take a bow at the same time.

4.3 Frobenius-like and étale-like

Every object in IUT comes in two flavours [IUTchIII, Rem. 1.5.4; Alien, §2.7]:

- **Frobenius-like** objects are monoids and Frobenioids that live inside one specific Hodge theater: the actual units, the actual theta values, the actual number field. They are rigid and precise, and they do *not* travel: they belong to one node of the lattice.
- **Étale-like** objects are rebuilt from Galois groups by anabelian algorithms. Because the Galois groups are coric, étale-like objects travel freely along any arrow that shares them. The price is that they are only defined up to the automorphisms of the Galois group.

The bridge between the two is **Kummer theory**: a Kummer isomorphism identifies a Frobenius-like monoid with its étale-like copy inside Galois cohomology. Mochizuki calls the act of crossing this bridge **Kummer-detachment** [Alien, §2.7 (vi)]. Crossing it costs an indeterminacy, and that cost is the subject of §7.

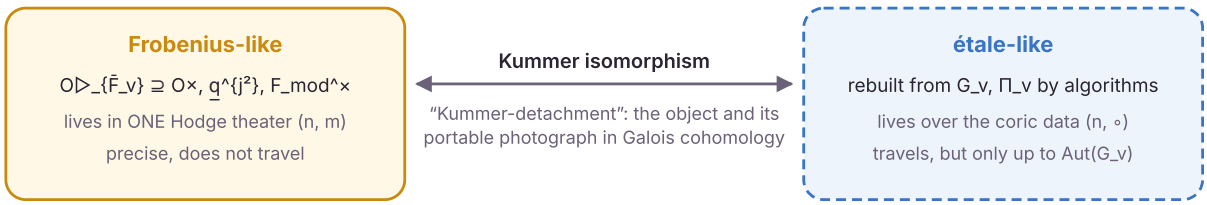


Figure 4.2. The two flavours of every object. The multiradial algorithm is a long sequence of crossings between them: detach the Θ-pilot from its Hodge theater into étale-like form, carry it along coric data, and land it in a container on the other side.

ANALOGY · THE CAKE AND THE PHOTO OF THE CAKE

A Frobenius-like object is a cake in your kitchen. An étale-like object is a photo of the cake, which you can send anywhere. The Kummer isomorphism says "this photo is of that cake". A photo can be received in another universe, but the receiver can rotate it, crop it, or print it at a different size; they cannot eat it. The multiradial algorithm is the art of sending enough photos, from enough angles, that the receiver can say how much cake there was, in their own units, to within a stated error.

5 The cast: every object the algorithm touches 登場人物紹介

A long-running series needs a character sheet. Here is everyone who appears in Theorem 3.11 and Corollary 3.12, in order of appearance, with where they are defined, what they do, and what they would be if IUT were an anime. The toolkit names (from `docs/IDENTIFIERS.md`, 448 identifiers in all) are given so you can find each one in `src/`, the web app and `lean/`.

character	defined in	role in the algorithm	if this were an anime
initial Θ-data	[IUTchI] Def 3.1	$(\overline{F}/F, X_F, l, \underline{C}_K, \underline{\mathbb{V}}, \mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}, \underline{\epsilon})$: the elliptic curve, the prime l, the places. Everything is built from it. `InitialThetaData`	the save file
Θ±ell NF-Hodge theater	[IUTchI] Def 6.13	one complete copy of arithmetic with labels $j \in \mathbb{F}_l$; one per lattice node. `ThetaPmEllNFHodgeTheater`	a parallel world
D-prime-strip, D⊢-prime-strip	[IUTchI] Def 4.1	the Galois groups $\{\Pi_v\}$ or $\{G_v\}$ alone, one per place: étale-like, coric. `DPrimeStrip`, `DMonoPrimeStrip`	the skeleton crew who appear in every world
F×μ-prime-strip	[IUTchII] Def 4.9	$\{G_v \curvearrowright \mathcal{O}_v^{\times\mu}\}$ plus realified global degrees: the gluing data of the Θ-link and the *input* of the algorithm. Called the BPS ("basic prime-strip") in [Form26]. `FMonoGlobValUnitsModTorsPrimeStrip`	the network cable between two computers [IUTchIII, Rem. 3.11.1 (iv)]
q-pilot, Θ-pilot	[IUTchIII] Def 3.8 (i)	$\underline{q}$ and $\{\underline{q}^{j^2}\}_j$ as objects of a global realified Frobenioid; the two things whose sizes are compared. `QPilot`, `ThetaPilot`	the two rival pilots in the final race
Θ×μ_LGP-link	[IUTchIII] Def 3.8 (ii)	full poly-isomorphism of F×μ-prime-strips from (n, m) to $(n+1, m)$, Θ-pilot $\mapsto$ q-pilot. `ThetaUnitsModTorsLGPTorsorLink`	the body swap
log-link	[IUTchIII] Def 1.1	the p-adic logarithm on units, from (n, m) to $(n, m+1)$. `LogLink`	the time leap within one world line
log-shell $\mathcal{I}_v$	[IUTchIII] Def 1.1	$\frac{1}{2p}\log(\mathcal{O}_{K_v}^{\times})$: the bi-coric additive container. `LogShell`	the stage on which everything is performed
procession $\mathrm{Prc}(\mathcal{D}_T)$	[IUTchI] Prop 6.9	$\{\mathcal{D}_0\} \to \{\mathcal{D}_0, \mathcal{D}_1\} \to \cdots \to \{\mathcal{D}_0, \ldots, \mathcal{D}_{l^*}\}$: labels added one at a time. `Procession`	a matsuri parade in which each float adds one more passenger
tensor packet $\mathcal{I}(\mathbb{S}^{\pm}_{j+1})$	[IUTchIII] Prop 3.2	tensor product of $j+1$ log-shells, one per label $0..j$. `TensorPacket`	a bento box with $j+1$ compartments
splitting / LGP-monoid $\Psi^{\perp}_{\mathrm{LGP}}$	[IUTchIII] Prop 3.4, 3.5	the theta values $\underline{q}^{j^2}$, acting multiplicatively on packet j. `PsiLGP`	the seasoning: one per box

copies of F_{mod}	[IUTchIII] Prop 3.7, 3.10	$M_{\mathrm{mod}}(\dots)_j$, one number field per label, from κ-coric functions; the global part. `FmodUnits`	the rice: the same in every box
étale theta function, mono-theta environment	[EtTh]; [IUTchII] §1–§2	its Kummer theory has a *rigid* cyclotome, so the theta values are insulated from (Ind1), (Ind2). `PsiTheta`	the one character whose sync ratio never drops
(Ind1), (Ind2), (Ind3)	[IUTchIII] Thm 3.11 (i), (ii)	the three blurs. `IndOne`, `IndTwo`, `IndThree`	the three-member comic relief
holomorphic hull	[IUTchIII] Rem 3.9.5	the smallest $\mathcal{O}$-module (arithmetic vector bundle) containing a region. `HolomorphicHull`	the furoshiki wrapping cloth
log-volume $\mu^{\log}$	[IUTchIII] Prop 3.9	the procession-normalized volume of a region: the scale both sides trust. `LogVolume`	the referee

OTAKU CORNER · OP CREDITS

The opening credits of IUT would run: *Original work* Shinichi Mochizuki. *Series composition* [IUTchI–IV]. *Character design* Grothendieck (posthumously). *Mechanical design* the étale theta function. *Background art* p-adic Hodge theory. *Music* the Gaussian integral. *Special thanks* Belyi, Faltings, Uchida, Tamagawa. Episode count: 646 pages, and the second cour is still being argued about on forums (§12).

6 The algorithm, step by step (Theorem 3.11) アルゴリズム完全解説

Here is the whole route on one page. The left column is Theorem 3.11 (build the multiradial representation); the right column is the proof of Corollary 3.12 (turn it into a number). The step labels (Stp1)–(Stp8) are those of [EssLgc] §3.10, which reorganizes the proof of Corollary 3.12 into eight moves; the lattice labels in brackets say where in Figure 4.1 each step happens.

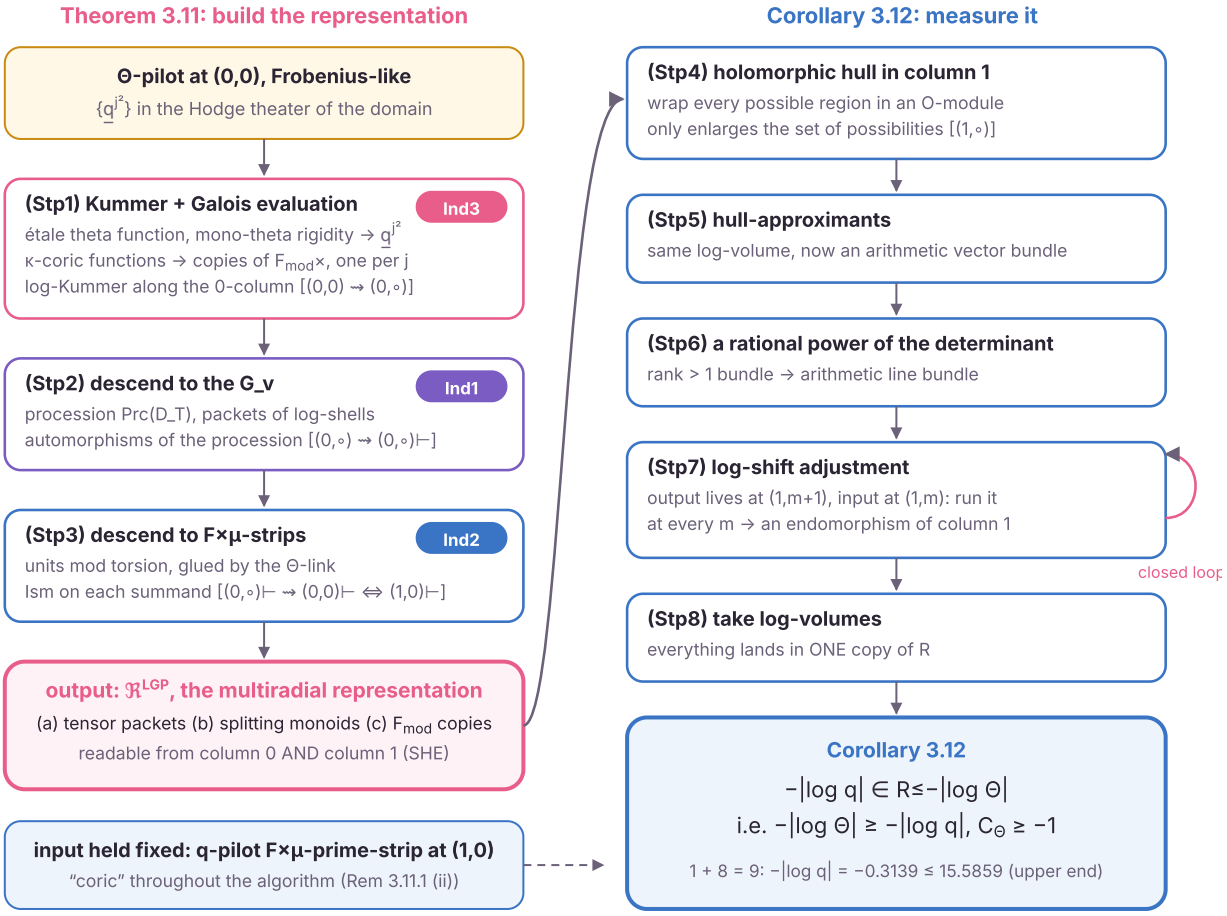


Figure 6.1. The multiradial algorithm and Corollary 3.12 as one flow. Coloured badges mark where each blur enters. In [Form26] Mochizuki regroups the same flow as "3.11 ⇒ 3.11.5 ⇒ 3.12", moving the hull and determinant (Stp4–Stp6) into an intermediate statement; see §8.4.

6.1 Step 0: the input is a prime-strip, and it is held fixed

A crucial reinterpretation comes first [IUTchIII, Rem. 3.11.1 (ii)]. Theorem 3.11 can be regarded as an algorithm "whose input data is an F×μ-prime-strip [...] and whose functoriality is with respect to arbitrary isomorphisms of the F×μ-prime-strips that appear as input data". Because the Θ-link identifies the F×μ-prime-strip of the Θ-pilot on the left with that of the q-pilot on the right, the input can be taken to be the **q-pilot's prime-strip at** $(1, 0)$, together with the ring structure of column 1 that rigidifies it. That input is then "coric": it "remains invariant"/"may be regarded as being held fixed" throughout the algorithm. Everything else moves; the q-pilot stays put. This is what makes the final comparison possible.

Step 1 (Stp1) · Kummer theory and Galois evaluation Ind3

in: the Frobenius-like Θ-pilot at $(0,0)$ · out: étale-like theta values and number fields over $(0,\circ)$ · [IUTchIII] Thm 3.11 (ii); [EssLgc] §3.10 (Stp1)

This is the step that occupies most of [IUTchI–III]. Its job is to exhibit the Frobenius-like Θ-pilot at $(0,0)$ "as one possibility within a collection of possibilities [...] constructed via anabelian algorithms from holomorphic [...] étale-like data" [EssLgc, §3.10 (Stp1)]. Three machines run in parallel:

- **Theta values.** The étale theta function on the Tate curve at each $v \in \mathbb{V}^{\mathrm{bad}}$ is reconstructed from the tempered fundamental group. Its Kummer theory runs through *mono-theta environments*, whose cyclotome is rigid ("mono-theta-theoretic cyclotomic rigidity", [EtTh]). Evaluating it at the l-torsion points labelled $j \in |\mathbb{F}_l|$ (Galois evaluation, synchronized across labels by the $\mathbb{F}_l^{\rtimes\pm}$-symmetry) produces the values $\underline{q}^{j^2}$ as étale-like monoids [IUTchII, Cor. 2.5, 3.5].
- **Number fields.** Kummer theory of κ-coric rational functions on the curve reconstructs copies $(F^{\times}_{\mathrm{mod}})_j$ of the multiplicative group of the field of moduli, one per label, related by the $\mathbb{F}_l^*$-symmetry [IUTchI, Ex. 5.1; IUTchII, Cor. 4.8].
- **The log-Kummer correspondence.** These Kummer isomorphisms exist at every node $(0,m)$ of the column, and Theorem 3.11 (ii) says they are compatible with the log-links, except that the images of the local units only satisfy inclusions: this is **(Ind3)**. A *non-interference* property keeps the damage small: the theta values and $F^{\times}_{\mathrm{mod}}$ meet the local units only in roots of unity [IUTchIII, Prop. 3.5 (ii)(c), 3.10 (ii)].

The companion *anabelian_guide.pdf* explains the reconstruction algorithms (Belyi and elliptic cuspidalization, Kummer theory, the three cyclotomic rigidities) that this step consumes.

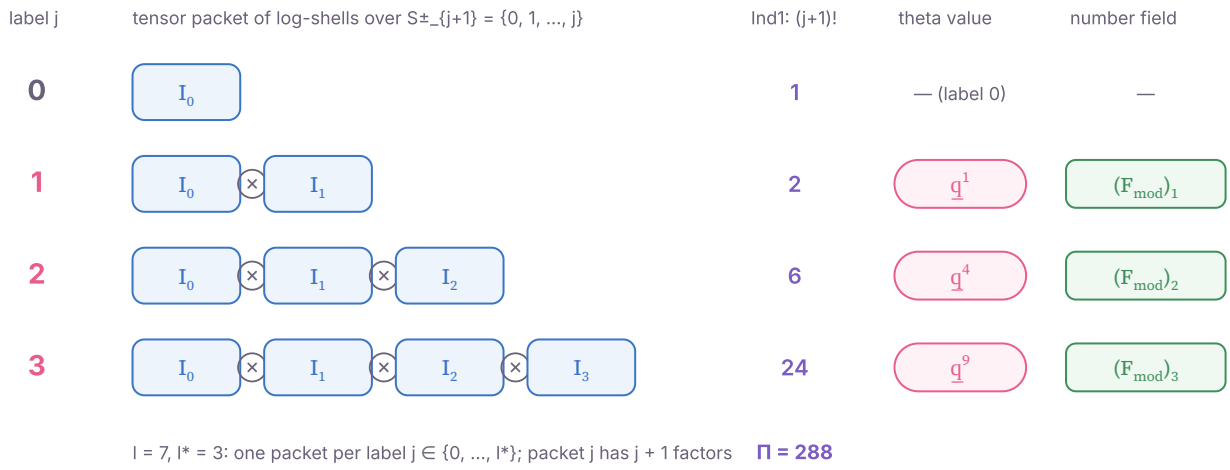


Figure 6.2. The data of Theorem 3.11 (i)(a)–(c) for $1+8=9$ ($l=7$). Row j is the tensor packet of $j+1$ log-shells over the procession container $\mathbb{S}^{\pm}_{j+1}=\{0,1,\ldots,j\}$, acted on by the theta value $\underline{q}^{j^2}$ (pink) and carrying a copy of F_{mod} (green). Purple numbers are the sizes $(j+1)!$ of the permutation groups that (Ind1) adds at each row. Compare [Alien] Fig. 3.18, "the full multiradial representation".

Concretely, at the bad place $v=3$ of the running example the Tate parameter has $\mathrm{ord}_3(q_v)=4$ and unit part 5110 modulo 3^8. The theta values the algorithm reconstructs are:

j	j^2	$\underline{q}^{j^2}$ at v = 3: $\mathrm{ord}_3(\underline{q}^{j^2})$	$\mathrm{ord}_3(q^{j^2})$	unit part mod 3^8	log-volume $-(j^2/2l)\cdot\log q$
1	1	2/7	4	5110	−0.3139
2	4	8/7	16	3130	−1.2556
3	9	18/7	36	3457	−2.8250

Step 2 (Stp2) · descend to the Galois groups Ind1

in: holomorphic étale-like data over $(0, \circ)$ · out: the same, rebuilt from the D⊢-prime-strips $\{G_v\}$ alone · [IUTchIII] Thm 3.11 (i); [EssLgc] §3.10 (Stp2)

The constructions of Step 1 used the full arithmetic fundamental groups Π_v. Step 2 observes that they "in fact descend to [...] algorithms that only require as input data the weaker data" of the *mono-analytic* Galois groups G_v. This is essential because G_v is what the Θ-link shares and Π_v is not. The data are arranged along the **procession** $\{\mathcal{D}_0\} \to \{\mathcal{D}_0, \mathcal{D}_1\} \to \cdots \to \{\mathcal{D}_0, \ldots, \mathcal{D}_{l^*}\}$: row j of Figure 6.2 is built over the capsule $\{\mathcal{D}_0, \ldots, \mathcal{D}_j\}$ and consists of

a. the **tensor packet** $\mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1})$: the tensor product of the $j+1$ log-shells, one per label, at each place $v_{\mathbb{Q}}$ of $\mathbb{Q}$ (a direct sum over the places v above $v_{\mathbb{Q}}$);
b. at $v \in \mathbb{V}^{\mathrm{bad}}$, the **splitting monoid** $\Psi^{\perp}_{\mathrm{LGP}}$, generated by $\underline{q}^{j^2}$, acting multiplicatively on the packets;
c. the **number field** $M_{\mathrm{mod}}(\ldots)_j \subseteq \mathcal{I}^{\mathbb{Q}}(\mathbb{S}^{\pm}_{j+1})$ with its global Frobenioids, whose global degrees are computed by log-volumes.

Working only with G_v introduces **(Ind1)**: the automorphisms of the procession. A capsule of $j+1$ unordered prime-strips has $(j+1)!$ automorphisms; the procession as a whole has $\prod_j (j+1)! = 288$ for $l = 7$ (and $24,883,200$ for $l = 11$). It also includes the automorphisms of each abstract G_v, which do not preserve the ring structure of the field it came from. Processions exist precisely to keep this blur small: without them the labels could be permuted in $(l^{\pm})^{l^{\pm}}$ ways, with them only $l^{\pm}!$.

TOOLKIT · IUT RUN COUNTFORGOTTENSYMMETRIES

```
CountForgottenSymmetries  [IUTchI] Prop 4.9, Prop 4.11, Prop 6.8, Prop 6.9 — (l*)^{l*} → l*! and (l±)^{l±} → l±!: what
the forgetful functors lose and what processions recover
  in: InitialThetaData   out: counts
    0NF (Prop 4.9, 4.11): F_l^* acts simply transitively on J (Prop 4.9 (i)); capsule of 3 strips, S_{l*} of order 6;
label assignments (l*)^{l*} = 27, with processions l*! = 6 (factor 4.5)
    0±ell (Prop 6.8, 6.9): F_l^{⋊±} acts doubly transitively on T (Prop 6.8 (i)); capsule of 4 strips, S_{l±} of order
24; (l±)^{l±} = 256, with processions l±! = 24 (factor 10.7)
    (ii) holomorphic capsules → (iii) mono-analytic capsules: the same counts after D ↦ D⊢
```

Step 3 (Stp3) · descend to the F×μ-prime-strip Ind2

in: data over $\{G_v\}$ · out: data over $\{G_v \curvearrowright \mathcal{O}_v^{\times\mu}\}$, which the Θ-link glues to $(1,0)$ · [IUTchIII] Thm 3.11 (i), (iii); [EssLgc] §3.10 (Stp3)

One more descent: the log-shells are now built "as submonoids of $\mathcal{O}_{F_v}^{\times\mu}$, as opposed to subquotients of G_v", i.e. from the *mono-analytic Frobenius-like* data of the F×μ-prime-strip. This is the data the Θ-link actually glues: the unit portion of ${}^{0,0}\mathcal{F}_{\mathrm{LGP}}^{\times\mu} \xrightarrow{\sim} {}^{1,0}\mathcal{F}^{\times\mu}$. Identifying Frobenius-like and étale-like units costs **(Ind2)**: independent copies of the isometry group Ism of $\mathcal{O}^{\times\mu}$ act on each direct summand of each factor of each tensor packet (for archimedean places, the automorphisms of order 2). The bi-coricity of the units (Theorem 1.5 (iii), (iv)) is what lets the mono-analytic log-shells serve as "multiradial mono-analytic containers" for the value-group data of column 0 [IUTchIII, proof of Cor. 3.12, Step (v)].

Why are the theta values not destroyed by (Ind1) and (Ind2)? Because their cyclotomes come from mono-theta environments and from F_{mod}, which are *insulated* from the automorphisms of $\mathcal{F}^{\times\mu}$ [IUTchIII, proof of Cor. 3.12, Step (ii), Fig. 3.6]. This is the single most important technical fact behind the theorem, and it is why the étale theta function, rather than any other function, sits at the centre of IUT.

6.2 The output

Write ${}^{n,\circ}\mathfrak{R}^{\mathrm{LGP}}$ for the data (a), (b), (c) regarded up to (Ind1) and (Ind2) (with (Ind3) recording how the Frobenius-like data at each (n,m) sit inside). Theorem 3.11 (i) says this can be constructed by an algorithm in the procession of D⊢-prime-strips, functorially; the permutation symmetries of the étale-picture give compatible isomorphisms ${}^{n,\circ}\mathfrak{R}^{\mathrm{LGP}} \xrightarrow{\sim} {}^{n',\circ}\mathfrak{R}^{\mathrm{LGP}}$ for all n, n'. In other words, **the representation built over column 0 is the same as the one built over column 1**. Theorem 3.11 (iii) adds that all of this is compatible with the Θ×μ_LGP-link: the constructions are "stabilized/equivariant/functorial with respect to arbitrary automorphisms of the domain and codomain" of the horizontal arrows.

TOOLKIT · IUT RUN REPRESENTMULTIRADIALLY

```
RepresentMultiradially  [IUTchIII] Thm 3.11 — the multiradial representation (a)(b)(c) with Ind1-Ind3
  in: LogThetaLattice   out: MultiradialRepresentation
    R^LGP_{0,◦}: (a) 4 packets, (b) 3 splitting monoids, (c) 3 copies of F_mod^×; Ind1 288, Ind2 on 4 summands, Ind3 1;
log-Kummer 1; link compat 1111
    −|log q| = -0.3139, −|log 0| ≤ 15.5859, C_0 = 31858372931939553247232.000, inequality: holds
```

(a) 4 tensor packets ($j = 0..3$), (b) 3 splitting monoids ($j = 1..3$), (c) 3 copies of $F_{\mathrm{mod}}^{\times}$; Ind1 = 1·2·6·24 = 288; Ind2 acts on the 4 summands; "link compat 1111" are the four compatibilities (a)–(d) of Thm 3.11 (iii).

6.3 The four properties: IPL, SHE, APT, HIS

Remark 3.11.1 names four properties of the output that the corollary uses. They are worth learning as a set, because they are what the rest of the argument actually consumes.

property	what it says	source
IPL · input prime-strip link	The output is linked, by full poly-isomorphisms of prime-strips produced by the algorithm itself, to the input: the q-pilot prime-strip with its column-1 ring structure. Every intermediate prime-strip could itself serve as input.	Rem 3.11.1 (iii)
SHE · simultaneous holomorphic expressibility	The construction and its output are "simultaneously valid/executable/well-defined" relative to *both* the ring structure of the domain (where the Θ-pilot lives) and that of the codomain (where the input lives).	Rem 3.11.1 (iii)

APT · algorithmic parallel transport	The transport from one side to the other is not a set-theoretic map of regions; it is an *algorithm* that can be run on either side. Its output regions cannot be compared directly with structures on the other side without further work (the hull, §8).	Rem 3.11.1 (iv)
HIS · hidden internal structures	For the corollary one may forget that the algorithm involves theta functions at all: it is "some" algorithm with input, output, IPL and SHE.	Rem 3.11.1 (iv)

WHAT THE PAPER SAYS · [IUTCHIII] REMARK 3.11.1 (IV)

"[T]heta functions/theta values may be regarded as HIS of the construction algorithm of Theorem 3.11 — somewhat like the internal structure of the CPU or operating system of a computer! — i.e., internal structures whose technical details are [...] irrelevant or uninteresting from the point of view of the 'end user' [...]. [T]he F×μ-prime-strips that occur in the Θ-link may be thought of as a sort [of] connecting cable, i.e., of the sort that is used to link distinct computers via the internet. That is to say, despite the fact that such a connecting cable may have a very simple internal structure by comparison to the computers that it connects, the connection that it furnishes has highly nontrivial" consequences.

OTAKU CORNER · ガチャ

(Ind1) is the gacha of IUT. For $l = 7$ there are 288 ways to arrange the procession and the algorithm must give the same log-volume for every pull. For $l = 11$ the banner has 24,883,200 outcomes, which is roughly the number of pulls it takes to get the limited-edition character you wanted, and the log-volume is still the same every time. Unlike a real gacha, the pity system is a theorem: Proposition 3.9 (ii) guarantees that procession-normalized log-volumes are invariant under (Ind1) and (Ind2). No whales were harmed in the making of this inequality.

OTAKU CORNER · シンクロ率

Mono-theta-theoretic cyclotomic rigidity is the IUT equivalent of a sync ratio that never drops. Every other cyclotome in the theory can be rescaled by $\widehat{\mathbb{Z}}^{\times}$ when you look at it from another universe, the way a pilot's synchronization wavers under stress. The cyclotome of the mono-theta environment cannot: its rigidity comes from the quadratic structure of theta-group commutators [IUTchIII, Rem. 2.1.1; Alien, §3.8 (9gau)], so it locks at one value whatever the universe. That single unflappable pilot is what lets the theta values cross the Θ-link intact.

7 The three blurs: (Ind1), (Ind2), (Ind3) 三つのぼかし

An indeterminacy is the price of a descent: whenever the algorithm replaces a rich object by a poorer one that travels better, it must accept every way the poorer object could have come from a rich one. The proof of Corollary 3.12 (Step (x)) explains where each of the three comes from in a single sentence each, and [Alien] §3.7 (i) gives a more concrete description. Side by side:

	(Ind1)	(Ind2)	(Ind3)
acts on	labels and Galois groups	units of each tensor factor	local units along the column
comes from	compatibility with the permutation symmetries of the étale-picture	compatibility with $\mathrm{Aut}(\mathcal{F}^{\times\mu})$, i.e. with the **horizontal** arrows	compatibility with the log-Kummer correspondence, i.e. with the **vertical** arrows

[Alien] name	étale-transport indeterminacy	Kummer-detachment indeterminacy (Ism)	Kummer-detachment indeterminacy (upper semi-commutativity)
group / shape	$\prod_j \mathfrak{S}_{j+1}$ and $\mathrm{Aut}(G_v)$; 288 label permutations here	Ism: a compact group of isometries of $\mathcal{O}^{\times\mu}$	not a group: inclusions $\subseteq$ (nonarch.), surjections $\twoheadrightarrow$ (arch.)
effect on log-volume	none (invariant)	none (invariant)	turns equality into an **upper bound**
enters at	(Stp2)	(Stp3)	(Stp1)

7.1 (Ind1): shuffled seats

Once the data are rebuilt from abstract Galois groups, nothing distinguishes one ordering of the capsule from another, and nothing pins down an isomorphism of G_v with the Galois group of a specific field. An automorphism of G_v need not come from a field automorphism (the absolute Galois group of a p-adic field does not determine the field; see *anabelian_guide.pdf*), so (Ind1) includes moves that scramble the ring structure. The log-volume does not see them: a procession-normalized log-volume is an average over labels and a sum over places, and both are symmetric.

7.2 (Ind2): turning the dial

Ism is the group of G_v-equivariant automorphisms of $\mathcal{O}_{\overline{k}}^{\times\mu}$ that preserve the lattices $\mathcal{O}_{k_H}^{\times\mu}$ for every open $H \subseteq G_v$ [IUTchII, Ex. 1.8 (iv)]. It is compact, so it moves points around while preserving volumes, the way rotating a dial preserves the area of the dial. Its existence is a genuine loss: the units on the two sides of the Θ-link are identified only up to such a twist, and in particular one cannot tell which unit is 1 + p and which is its image under some isometry. Each tensor factor gets its own independent copy, so the twist cannot even be synchronized across labels.

7.3 (Ind3): only "fits inside"

Along the 0-column the Kummer isomorphisms at different heights m are compatible with the log-links only in the sense that the log-shell at one level *contains* the image of the units from the next. Mochizuki calls this **upper semi-compatibility** (Theorem 3.11 (ii), Prop. 3.5 (ii)). It is the price of the "vertical shift" that the argument needs: the units must be taken one level below the value groups they measure, and vertically coric containers are the only way to accommodate that [IUTchIII, proof of Cor. 3.12, Step (iv)]. In terms of log-volume, an inclusion can only make things bigger, so (Ind3) converts the would-be equality into an inequality from above. The log-volumes themselves are compatible with the log-links on the nose (Prop. 3.9 (iv)).

WHY THE BLURS ARE HARMLESS FOR LOG-VOLUMES · [IUTCHIII] PROOF OF COR 3.12, STEP (X)

The three indeterminacies describe "some sort of 'formal quotient', like the 'fine moduli spaces' that appear in algebraic geometry. [...] [T]he procession-normalized mono-analytic log-volumes [...] furnish a means of constructing a sort of associated 'coarse space' [...] in the sense that [...] the resulting log-volumes $\in \mathbb{R}$ are invariant with respect to the indeterminacies (Ind1), (Ind2), and have the effect of converting the indeterminacy (Ind3) into an inequality [from above]."

7.4 The invisible indeterminacies

Two more blurs exist but can be ignored "essentially as a formal consequence of the way in which the various objects that appear are defined" [Alien, §3.7 (i); IUTchIII, Rem. 3.11.4]: multiplication by roots of unity (the only overlap between theta values and local units, by non-interference, and the theta values

are stable under it), and the sign ± 1 in the cyclotomic rigidity for κ-coric functions (the global Frobenioids only use all of $F^{\times}_{\mathrm{mod}}$, which is stable under inversion).

7.5 Watching the blur on real numbers

The toolkit's `Fuzzify` process implements the "engulfing" picture of [EssLgc] §3.10: start from the Θ-pilot's naive log-volume and widen it by everything the three indeterminacies and the hull allow, then ask whether the q-pilot is inside.

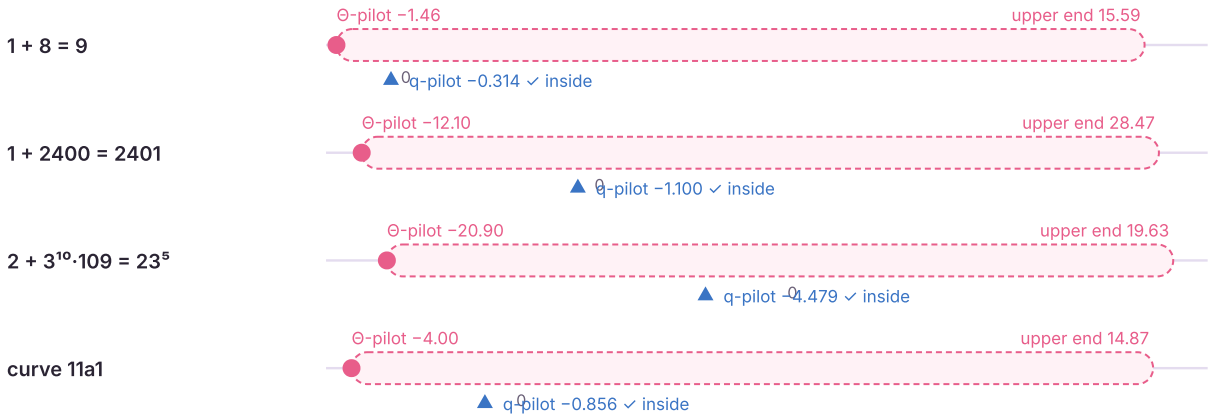


Figure 7.1. For each worked example: the Θ-pilot's naive log-volume (pink dot), the range of log-volumes the blurred and hulled representation allows (pink capsule, up to its upper end), and the q-pilot (blue triangle). In every case the q-pilot is engulfed, as Corollary 3.12 predicts. Each row has its own scale.

TOOLKIT · IUT RUN FUZZIFY

```
Fuzzify  [EssLgc] §3.10 (Englf), (ExtInd1/2), §3.11 — fuzzification, engulfing, extending indeterminacies, ∨ verdicts
  in: value, indeterminacies   out: interval
    0-pilot -1.4648 fuzzified by Ind1-3 to [-1.4648, 15.5859]: engulfs −|log q| = -0.3139: yes;  q-pilot fuzzified sym-
metrically: [-1.3139, 17.7369]
    (ExtInd1) {1,2} ∪ {2,3} has 3 elements; (ExtInd2) crushed by |·|: 3 classes; (logORInd): harmless — log-volume is
invariant under the log-link; (ΘORInd): fatal — an OR across the Θ-link asserts nothing
```

"(ExtInd1/2)" illustrate the two ways [EssLgc] says possibilities may be added (unions of sets of possibilities, and passing to coarser classes); "(logORInd)" and "(ΘORInd)" are its point that an "OR" across a log-link is harmless because log-volumes are log-link invariant, while an "OR" across the Θ-link would say nothing at all (§10).

ANALOGY · SHINKANSEN RESERVED SEATING

Think of the procession as a Shinkansen car with reserved seats. (Ind1): the conductor reshuffles who sits in which seat. (Ind2): every passenger rotates in their seat (they can, the seats swivel). (Ind3): the luggage rack is "at least as big as" the luggage, not exactly as big. Ask how much the car weighs: the first two do not change the answer, and the third means the scale can only say "at most this much". The train still leaves on time, to the second.

OTAKU CORNER · 三人組

Every long-running comedy has a trio of troublemakers who undo the protagonist's careful plans. IUT's trio are (Ind1), (Ind2) and (Ind3). Mochizuki insists they are "relatively mild"; readers have long asked whether one of them is quietly carrying the plot. [EssLgc] §3.11 even ranks them, "(Ind3 > 1 + 2)": once the output is stable under (Ind3), stability under the other two comes almost for free. The equivalent in a shōnen manga is the reveal that the quiet one was the strongest all along.

8 From picture to inequality (Corollary 3.12) 絵から不等式へ

COROLLARY 3.12 · LOG-VOLUME ESTIMATES FOR Θ-PILOT OBJECTS

Let $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R} \cup \{+\infty\}$ be the procession-normalized mono-analytic log-volume of the **holomorphic hull** of the union of the possible images of a Θ-pilot object in the multiradial representation of Theorem 3.11, subject to (Ind1), (Ind2), (Ind3). Let $-|\log(\underline{q})| \in \mathbb{R}$ be the procession-normalized log-volume of the image of a q-pilot object, *not* subject to the indeterminacies. Then $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R}$ and

$$-|\log(\underline{\underline{\Theta}})| \;\geq\; -|\log(\underline{q})|,$$

i.e. $C_\Theta \geq -1$ for any real C_Θ with $-|\log(\underline{\underline{\Theta}})| \leq C_\Theta \cdot |\log(\underline{q})|$.

Two asymmetries in the statement are the whole point. The Θ-side is *blurred and hulled*; the q-side is *sharp*. And the inequality goes the "surprising" way: naively the Θ-pilot is the bigger object (its log-volume is more negative, -1.4648 against -0.3139 for $1+8=9$), yet the corollary says the blurred Θ-pilot cannot be smaller than the q-pilot. The only way both can be true is if the blur is at least as big as the gap. That is the height bound in disguise (§9).

8.1 The proof, in Mochizuki's twelve steps

The proof begins with a free move: since $|\log \underline{q}| > 0$, one may assume $-|\log \Theta| < 0$, because otherwise $-|\log \Theta| \geq 0 > -|\log \underline{q}|$ and there is nothing to prove. Then it reviews "precisely what is achieved" by the theory, in steps:

step	content
(i)	Concentrate on a **single** Θ×μ_LGP-link $(0,0) \to (1,0)$. It has a unit portion ($\mathcal{F}^{\times\mu}$) and a value-group portion ($\mathcal{F}^{\shortparallel}$); the latter sends the Θ-pilot to the q-pilot.
(ii)	Units suffer $\mathrm{Aut}_{\mathcal{F}^{\times\mu}}$ (that is, (Ind1), (Ind2)); the cyclotomes behind the value-group portion (mono-theta environments, κ-coric functions) are **insulated** from them.
(iii)	Units and value groups must be related through the *same* Θ-link, because the lattice does not commute.
(iv)	But the units must sit one level below the value groups ("vertical shift"). Resolve the tension with vertically coric structures, at the cost of (Ind3).
(v)–(ix)	Use bi-coric mono-analytic log-shells as containers; recall how the log-Kummer correspondences, conjugate synchronization by $\mathbb{F}_l^{\times\pm}$, the $\mathbb{F}_l^*$-symmetry and the copies of F_{mod} make this work. Three approaches to cyclotomic rigidity, all compatible (Fig. 3.7 of [IUTchIII]).
(x)	Summary: Kummer-detachment relates the Frobenius-like data at $(0,0)$ to the multiradial representation at the cost of (Ind1)–(Ind3); log-volumes are the "coarse space" that forgets them.
(xi-a)–(xi-g)	The comparison itself: see §8.2.
(xii)	Why **global** realified Frobenioids are needed: local Frobenioid objects have non-invertible endomorphisms, so locally one could not tell a region from its p^N-multiple.

8.2 Substeps (xi-a)–(xi-g): two ways to compute one number

1. **(xi-a)** Think of the q-pilot at $(1,0)$ in terms of the holomorphic log-shells there. The Θ-link is a gluing that makes the Θ-pilot at $(0,0)$ correspond to the q-pilot at $(1,0)$.
2. **(xi-b)** The multiradial algorithm produces a collection of possible output data in ${}^{0,\circ}\mathcal{U} \xrightarrow{\sim} {}^{1,\circ}\mathcal{U}$, the arrow coming from the étale-picture symmetry. For the rest of the argument, forget the theta functions (HIS) and keep only IPL and SHE.

3. **(xi-c)** So: the output possibilities are linked to the q-pilot's representation on ${}^{1,\circ}\mathcal{U}^{\mathbb{Q}}$ and can be expressed entirely in the ring structure of column 1. Pass to the **holomorphic hull**: the possibilities become localizations of arithmetic vector bundles over the rings $\mathcal{O}_{K_v}$ labelled $(1, \circ)$.
4. **(xi-d)** Those bundles have rank > 1 and the q-pilot is a line bundle, so take (a power of) the determinant and then the log-volume. Two comparable objects result: $\mathbb{R}_{\leq -|\log \Theta|}$ and $-|\log \underline{q}| \in \mathbb{R}$. The log-Kummer correspondence in column 1 corrects the vertical shift between the units of ${}^{1,0}\mathcal{F}^{\times\mu}$ and the log-shells built from them.
5. **(xi-e)** Expressibility in column 1's ring structure means: valid even under the condition that the input prime-strip's pilot log-volume equals the fixed value $-|\log \underline{q}|$.
6. **(xi-f)** Hence the construction of $\mathbb{R}_{\leq -|\log \Theta|}$ "constitutes, in particular, a construction [...] of the pilot-object log-volume of the input data", namely $-|\log \underline{q}|$. So $-|\log \underline{q}| \in \mathbb{R}_{\leq -|\log \Theta|}$, and the inequality "follows formally".
7. **(xi-g)** In summary: two tautologically equivalent ways to compute the log-volume of the q-pilot at $(1, 0)$ (Figure 8.1).

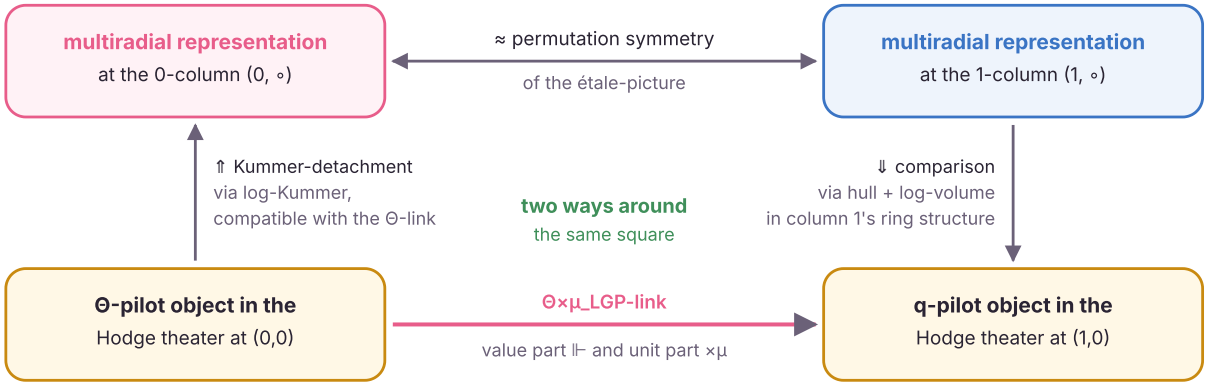


Figure 8.1. After [IUTchIII] Fig. 3.8, "two tautologically equivalent ways to compute the log-volume of the q-pilot object at $(1, 0)$". Bottom route: the Θ-link says the q-pilot *is* (the image of) the Θ-pilot. Top route: the Θ-pilot, detached into the multiradial representation, transported across by the étale-picture symmetry, hulled and measured in column 1. Both routes compute the same thing, so the measured value must be compatible with the q-pilot's own: that compatibility is the inequality.

THE PUNCHLINE IN ONE LINE

The set of log-volumes the algorithm can output, $\mathbb{R}_{\leq -|\log \Theta|}$, was constructed by an algorithm that, run on the q-pilot's own input, is a way of computing the q-pilot's log-volume. So the q-pilot's log-volume is one of the outputs: $-|\log \underline{q}| \leq -|\log \Theta|$.

8.3 The hull, the determinant and the closed loop (Stp4–Stp8)

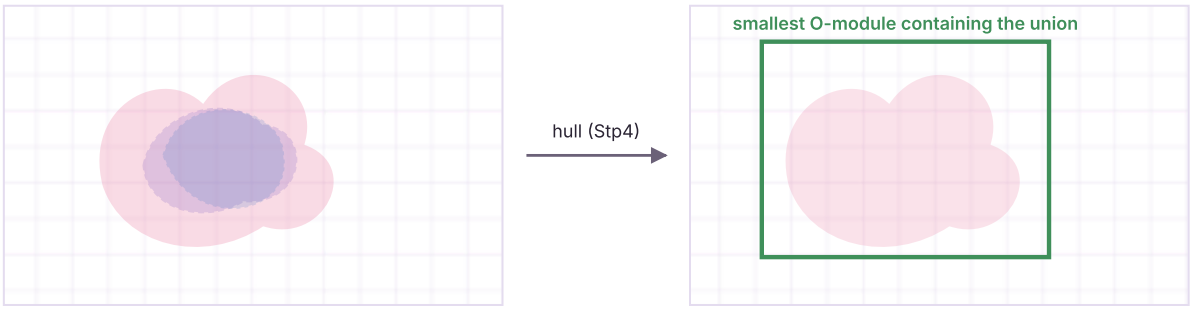


Figure 8.2. The holomorphic hull. The possible images of the Θ-pilot are arbitrary regions in the tensor packets (left); the q-pilot is an arithmetic line bundle, so the two cannot be compared until the regions are wrapped in the smallest module over the ring of integers that contains them all (right). Wrapping only enlarges, which is another reason the result is an upper bound.

In [EssLgc] §3.10 the same substeps appear as five moves, each of which "does not affect the logical '∧/∨' structure" or only adds possibilities:

- **(Stp4) Hull.** Pass from the possible regions to the holomorphic hull in column 1: a "simple, straightforward" enlargement of the set of possibilities.
- **(Stp5) Hull-approximants.** Replace each region by one with the same global log-volume that corresponds to an arithmetic vector bundle.
- **(Stp6) Determinant.** Pass to a suitable rational tensor power of the determinant line bundle.
- **(Stp7) Log-shift adjustment.** The line bundles of (Stp6) live in the ring structure at $(1,1)$, one step above the Θ-link's codomain. Running (Stp1)–(Stp6) at every $(1,m)$ gives arrows from $(1,m)$ to $(1,m+1)$: "a sort of endomorphism of the 1-column", a **closed loop**. Because log-volumes are compatible with log-links, the shift costs nothing.
- **(Stp8) Log-volumes.** The closed loop means the logical AND relation kept through (Stp1)–(Stp7) becomes an AND relation between the q-pilot's log-volume and the set of possible output log-volumes "within the same container, i.e., some copy of the real numbers". The inequality follows formally.

TOOLKIT · IUT RUN CHAINANDSTEPS

```
ChainAndSteps  [EssLgc] §3.10 (Stp1)-(Stp8), (FxGl) — the chain of logical ∧ relations through the lattice
  in: LogThetaLattice   out: steps with log-volumes
    StpOne    (0,0)        ↠ (0,◦)                Ind3         15.5859   [EssLgc] §3.10 (Stp1); [IUTchIII] Thm 3.11
(ii)
    StpTwo    (0,◦)      ↠ (0,◦)⊢             Ind1         15.5859   [EssLgc] §3.10 (Stp2); [IUTchIII] Thm 3.11 (i)
    StpThree  (0,◦)⊢   ↠ (0,0)⊢ ⇔ (1,0)⊢  Ind2         15.5859   [EssLgc] §3.10 (Stp3); [IUTchII] Cor 4.10 (iv)
    StpFour   (1,0)⊢     ↠ (1,◦) hull           none         17.0507   [EssLgc] §3.10 (Stp4); [IUTchIII] Cor 3.12 proof
    StpFive   (1,◦) hull ↠ hull-approximants      none         17.0507   [EssLgc] §3.10 (Stp5)
    StpSix    approximants ↠ determinant^{1/n}      none         17.0507   [EssLgc] §3.10 (Stp6)
    StpSeven  (1,m)↑     ↠ (1,m+1)↓             none         17.0507   [EssLgc] §3.10 (Stp7); [IUTchIII] Prop 1.2 (iii)
    StpEight  log-volumes  ↠ R: −|log 0| ≥ −|log q| monodromy    -0.3139   [EssLgc] §3.10 (Stp8); [IUTchIII] Cor 3.12
    (FxGl) q-pilot fixed at (1,0): -0.3139; glued 0-pilot at (0,0): -1.4648
```

The last column is the log-volume after each step for $1+8=9$. (Stp1)–(Stp3) carry the procession-normalized value 15.5859; the hull (Stp4) raises it to 17.0507 before the theta shift is subtracted; (Stp8) lands at the q-pilot's -0.3139 with the "monodromy" of the closed loop.

TOOLKIT · IUT RUN ESTIMATELOGVOLUME

```
EstimateLogVolume  [IUTchIII] Cor 3.12 — −|log Θ| ≥ −|log q| as numbered steps
  in: LogThetaLattice   out: numbers
    (0) |log q| = log q / 2l: log-volume of the q-pilot at (1,0), i.e. −|log q|                  -0.3139   [IUTchIII]
Cor 3.12; Def 3.8 (i)
    (1) the Θ-pilot at (0,0): naive log-volume −avg(j²)·|log q|                                -1.4648   [IUTchIII] Def
3.8 (i)
    (2) the Θ×μ_LGP-link (0,0)→(1,0) identifies the Θ-pilot with the q-pilot (value groups), units up to Ism
-0.3139   [IUTchIII] Def 3.8 (ii), Thm 3.11 (iii)
    (3) multiradial representation of the Θ-pilot in terms of the mono-analytic data of (1,0): Ind1 count      24.0000
[IUTchIII] Thm 3.11 (i)
    (4) holomorphic hull of the region: Σ_j (j+1) log d^K (procession-normalized)                17.0507   [IUTchIII]
Cor 3.12 proof; [Alien] §3.7 (iv)
    (5) −|log Θ| upper bound (hull + theta shift, procession-normalized)                         15.5859   [IUTchIII]
Cor 3.12
    (6) inequality −|log Θ| ≥ −|log q| ⇔ C_Θ ≥ −1: C_Θ                  318583729319395532472320.0000
[IUTchIII] Cor 3.12; [IUTchIV] Thm 1.10
    Cor 3.12 holds: yes
```

8.4 The 2026 reorganization: 3.11 ⇒ 3.11.5 ⇒ 3.12

For the Lean formalization Mochizuki regroups the argument [Form26, §4]. A diagram of *species* (types of objects) and *mutations* (functorial algorithms) has four commuting triangles. The first two express the multiradial algorithm, with input a BPS (basic prime-strip, i.e. F×μ-prime-strip), as a **descent** (dsc): it only needs a BPS glued to the vertically coric étale Hodge theater (etHT). The third composes it with "hull+det" to get (HDD). The fourth restricts (HDD) along the SHE arrow to the case where the input is the q-pilot. The never-stated "Theorem 3.11.5 (= 3.11 + Rmk. 3.9.5)" is the statement after the third triangle, and "3.11.5 ⇒ 3.12" is the simultaneous comparison of Θ-pilot and q-pilot in one ring structure. Skeletal Lean code exists for this last step; work on the second and third triangles (the APT property) is under way at RIMS.

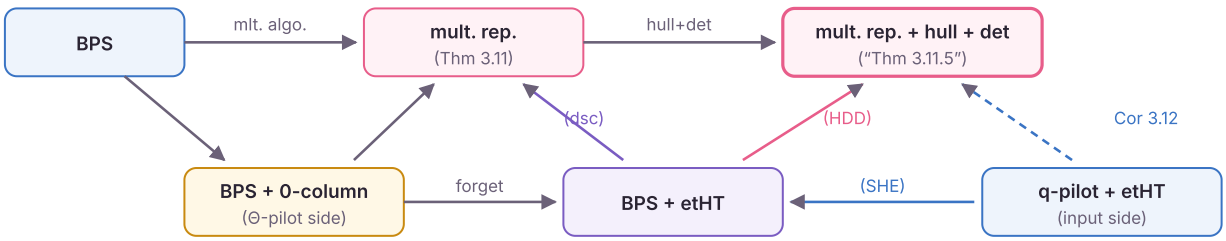


Figure 8.3. A simplified redrawing of the species/mutation diagram of [Form26] §4. (dsc) is the descent arrow; (HDD) := (hull+det) ∘ (dsc); restricting (HDD) along (SHE) to the q-pilot input and comparing with the q-pilot itself is "3.11.5 ⇒ 3.12". The toolkit's Lean port has a toy version in `lean/IUT/LeanForm.lean`.

8.5 Three "why" questions the proof answers

Why only one Θ-link? Because the lattice does not commute. Relating units through one Θ-link and value groups through another would need a commuting square (Step (iii)).

Why global Frobenioids? The Θ-link only preserves isomorphism classes of objects. Locally, a region and its p^N-multiple are related by endomorphisms that are not automorphisms, so a purely local argument could not even tell them apart. Global realified Frobenioids have no such endomorphisms (Step (xii)).

Why not use Θ^N and get a sharper bound? Because the argument rests on the first power of the étale theta function: its mono-theta cyclotomic rigidity has no evident analogue for N-th tensor powers, whose log-volume "must always be computed as the result of multiplying the log-volume of the original Θ-pilot object by N". Mochizuki adds that the sharper inequalities this would give "are known to be

false” (Step (xi-h), [IUTchIV] Rem. 2.3.2 (ii)). On the other hand, replacing the q-pilot by $\underline{q}^{\lambda}$ on the codomain side is harmless and gives $C_{\Theta} \geq -\lambda$ (Rem. 3.12.1 (ii)).

OTAKU CORNER · 風呂敷

The holomorphic hull is a furoshiki. You have a pile of oddly shaped objects (the possible regions); to carry them on the train you wrap them in one square cloth, tied at the corners. The cloth is bigger than the objects and its size is easy to state (“a 70 cm furoshiki”), which is the point. Corollary 3.12 is the statement that the q-pilot fits in the cloth. A sceptic might ask whether the cloth is so big that anything would fit; §9 shows how big it is for real curves, and §12 is where people argue about it.

9 The numbers: four worked examples 実際の数値

The toolkit's `iut pipeline` runs the whole theory on exact initial Θ-data for four elliptic curves (three Frey curves of abc triples and the curve 11a1; see `docs/PIPELINE.md`). Stage 5 of the pipeline is the multi-radial representation and Corollary 3.12. Here is what it produces.

example	l	V^{bad}_{mod}	log q	−|log q| (q-pilot)	Θ-pilot	Ind1 = Π(j+1)!	hull	upper end for −|log Θ|	Cor 3.12
1 + 8 = 9	7	{3}	4.394	−0.3139	−1.4648	288	17.051	15.586	√
1 + 2400 = 2401	11	{3, 5, 7}	24.202	−1.1001	−12.1011	24,883,200	40.572	28.470	√
$2 + 3^{10}\cdot 109 = 23^5$	7	{3, 23, 109}	62.710	−4.4793	−20.9033	288	40.531	19.628	√
curve 11a1	7	{11}	11.989	−0.8564	−3.9965	288	18.869	14.873	√

Columns: the bad places of F_{mod} used for $\log q$ (odd primes only, [IUTchI] Def 3.1); the q-pilot $-|\log q| = -\log q/2l$; the Θ-pilot $-\frac{l(l+1)}{12}|\log q|$; the size of (Ind1); the hull term $\frac{1}{l^*}\sum_{j=1}^{l^*}(j+1)\log d^K$; the upper end of the blurred Θ-pilot after the theta shift; whether $-|\log q| \le$ that upper end.

9.1 Row by row, for $1 + 8 = 9$

The algorithm's output, measured, is a sum over the procession. Packet j contributes the hull of $j+1$ log-shells, which costs $(j+1)\log d^K$ (the different of K measures how big a log-shell is), and the theta value $\underline{q}^{j^2}$ acting on it, which contributes $-\frac{j^2}{2l}\log q$ [Alien, §3.7 (iv); IUTchIV, Thm 1.10, Step (v)].

j	$\lvert S^{\pm}_{j+1}\rvert$	(j+1)!	(j+1)·log d^K	−(j²/2l)·log q	row total
0	1	1	5.6836	0.0000	**5.6836**
1	2	2	11.3672	−0.3139	**11.0533**
2	3	6	17.0507	−1.2556	**15.7952**
3	4	24	22.7343	−2.8250	**19.9093**

Averaging the rows $j = 1, \ldots, l^*$ gives the upper end 15.5859: that is the hull term 17.0507 plus the Θ-pilot's naive log-volume -1.4648. The q-pilot sits at -0.3139, far below, so the inequality holds by a wide margin.

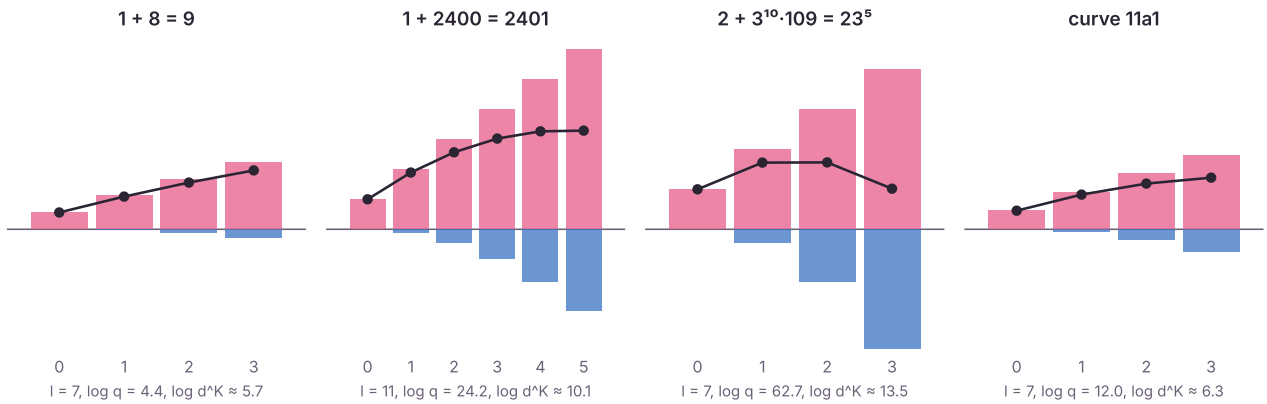


Figure 9.1. Per-label contributions for all four examples, on a common scale: the hull cost $(j+1)\log d^K$ (pink, up), the theta shift $-\frac{j^2}{2l}\log q$ (blue, down), and their sum (black). The hull grows linearly in j, the theta shift quadratically. For $2 + 3^{10}\cdot 109 = 23^5$, whose $\log q$ is large relative to $\log d^K$, the parabola already turns over at $j = 3$.

9.2 From the corollary to a height bound

Write the measured upper end in closed form. With $\sum_{j=1}^{l^*}(j+1) = \frac{l^*(l^*+3)}{2}$ and $\sum_{j=1}^{l^*} j^2 = \frac{l^*\,l\,(l+1)}{12}$ (both proved in `lean/IUT/Multiradial.lean`),

$$-|\log\Theta| \;\lesssim\; \frac{1}{l^*}\sum_{j=1}^{l^*}\Big[(j+1)\log d^K - \frac{j^2}{2l}\log q\Big] \;=\; \frac{l+5}{4}\log d^K \;-\; \frac{l+1}{24}\log q.$$

Corollary 3.12 says this is at least $-|\log q| = -\frac{\log q}{2l}$. Rearranging,

$$\log q\left(\frac{l+1}{24} - \frac{1}{2l}\right) \;\le\; \frac{l+5}{4}\log d^K \qquad\Longleftrightarrow\qquad \log q \;\le\; \frac{6\,l\,(l+5)}{l^2+l-12}\log d^K.$$

For large l the factor tends to 6, so the inequality reads $\frac{1}{6}\log q \lesssim (1+O(1/l))\log d^K$. The left side is the height of the curve (the toolkit's $\mathrm{ht}_E = \frac{1}{6}\log q$); the right side is controlled by the conductor and the different. That is a Szpiro-type inequality, and through it abc. Before normalizing by l^*, the same computation is Mochizuki's "leading term" $\frac{l^2}{48}\big(6\log d^K - \log q\big)$ [Alien, §3.7 (iv)]: "a 'large positive constant' $\frac{l^2}{48}$ times precisely the quantity [...] that one wishes to bound from below". He notes that this computation "was in fact known to the author around the year 2000", and that a primary motivation for IUT was "establishing a suitable framework, or geometry, in which this computation could be performed."

example	l	6l(l+5)/(l²+l−12)	log d^K (estimate)	bound on log q	actual log q	log q / 6 log d^K
1 + 8 = 9	7	11.455	5.684	65.103	4.394 √	12.9 %
1 + 2400 = 2401	11	8.800	10.143	89.258	24.202 √	39.8 %
2 + 3^10·109 = 23^5	7	11.455	13.510	154.756	62.710 √	77.4 %
curve 11a1	7	11.455	6.290	72.046	11.989 √	31.8 %

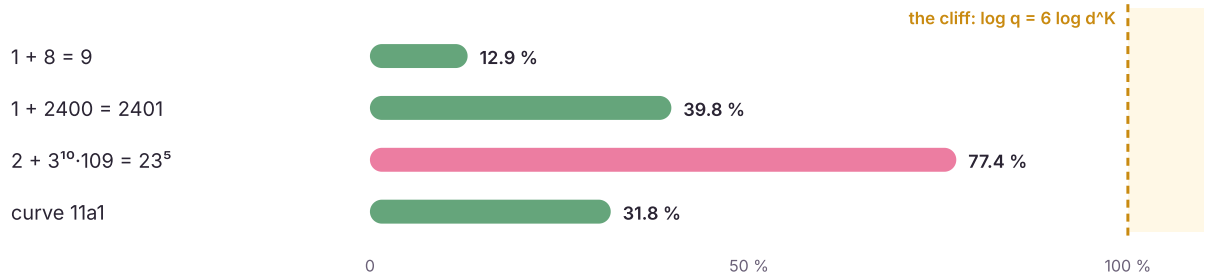


Figure 9.2. How close each example comes to the edge $\log q = 6\log d^K$, where the leading term of the multiradial representation changes sign. The record abc triple $2 + 3^{10}\cdot 109 = 23^5$ gets furthest, as it should: a high-quality abc triple is exactly a curve with large $\log q$ for its conductor.

9.3 The constant C_Θ and Theorem 1.10

[IUTchIV] Theorem 1.10 replaces the leading-term heuristic by a fully explicit estimate, including the archimedean places, the places above 2, and all the error terms, and expresses the result as an upper bound C_Θ with $-|\log\Theta| \le C_\Theta\,|\log q|$. The toolkit evaluates it: for $1+8=9$ it gets $C_\Theta = 3.19\cdot 10^{22}$, comfortably ≥ -1, and Theorem 1.10's height inequality reads $\frac{1}{6}\log q = 0.7324 \le 1.00\cdot 10^{22}$.

HOW TO READ THESE NUMBERS

- **The constants dominate.** Theorem 1.10's right-hand side contains $\eta_{\mathrm{prm}} = 5 \cdot 10^{20}$ ([ExpEst] §1). For curves this small it swamps everything, so the bound holds trivially and says nothing sharp. The theory is about curves of enormous height, where l is chosen near $\sqrt{\mathrm{height}}$ [IUTchIV, Cor. 2.2].
- $\log d^K$ **is an estimate.** The initial Θ-data files record no ramification data for $K = F(E[l])$, so the toolkit estimates the different and takes $e = 1$ in the log-shells.
- **Only** $\log q$ **is exact.** It is computed from the Tate parameters at the odd bad places and cross-checked against an independent computation (stage 7 of the pipeline).
- **The toolkit does not prove anything.** It implements the data types and the bookkeeping of the statements. Whether the statements are proved is the subject of §12.

OTAKU CORNER · 体重測定

Sumo wrestlers are weighed before each tournament, and the number is public. Corollary 3.12 is a weigh-in where one wrestler (the Θ-pilot) steps on the scale wearing a heavy keshō-mawashi (the hull) and the other (the q-pilot) in a plain mawashi. The rule says the dressed-up wrestler must weigh at least as much as the plain one. For our four small examples the keshō-mawashi weighs several times more than either wrestler, which is why the rule is satisfied easily. In the big tournaments, curves with height in the millions, the apron is relatively light and the rule starts to bite.

10 Logic: why "AND" and not "OR" 論理：「かつ」と「または」

[EssLgc] is Mochizuki's 2022 paper on the logical structure of the theory. Its central claim is that the whole argument is a chain of logical AND relations ($\wedge$), and that reading any link of the chain as an OR ($\vee$) makes the theory look either trivial or wrong. Example 2.4.5 gives a model with real numbers $A, B > 0$ and $0 \leq \varepsilon \leq 1$ that mirrors the three acts of §1.3:

act	AND version (IUT)	OR version
Θ-link	$N \stackrel{\text{def}}{=} -2B \ \wedge \ N \stackrel{\text{def}}{=} -A$	$N = -2B \ \vee \ N = -A$
multiradial representation	$N = -2A + \varepsilon \ \wedge \ N = -A$	$N = -2A + \varepsilon \ \vee \ N = -A$
final estimate	$-2A + \varepsilon = -A$, so $A = \varepsilon \leq 1$	nothing follows

Read the dictionary: N is the glued object, $-A$ the q-pilot's log-volume, $-2B$ the Θ-pilot's in its own universe (a dilation by 2 standing in for j^2), and $-2A + \varepsilon$ the multiradial representation of the Θ-pilot, computed in the q-pilot's universe up to a small error ε. In the AND reading the Θ-link is consistent precisely because A and B are allowed to be different numbers (two universes). The multiradial representation then re-expresses $-2B$ in terms of A, and both equations hold at once, which forces $A \leq 1$: a bound on the height.

In the OR reading the distinct A, B look superfluous, one is tempted to set $A = B$ "for the sake of simplicity", at which point the AND version of the Θ-link becomes self-contradictory ($-2A = -A$) and the passage to the final estimate "looks as if one erroneously replaced the '$\vee$' [...] by an '$\wedge$' without any mathematical justification whatsoever." Mochizuki attributes the 2018 objection (§12) to this chain, which he calls the "redundant copies school" (RCS).

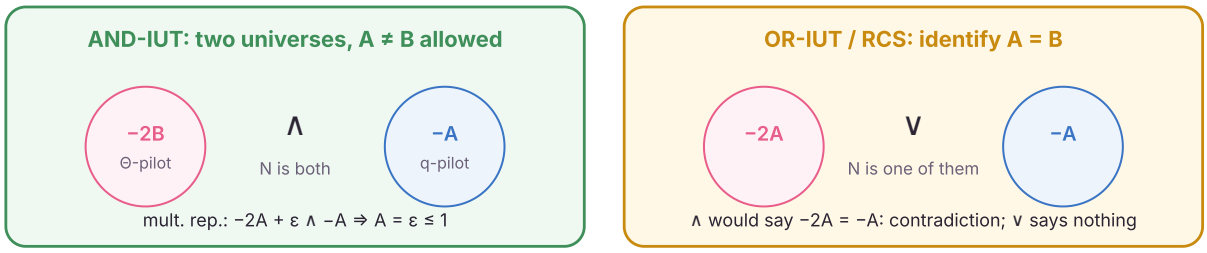


Figure 10.1. [EssLgc] Example 2.4.5 as a picture. The AND reading needs two universes; the OR reading collapses them and loses the content.

TOOLKIT · IUT RUN EVALUATELOGICMODEL

```
EvaluateLogicModel  [EssLgc] Ex 2.4.5 — the numerical model: ∧ gives A = ε ≤ 1, ∨ gives nothing
  in: A, B, ε   out: conclusion
    AND-IUT (∧)                0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ A = ε ≤ 1
    OR-IUT (∨)                 0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
    XOR-IUT                      0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
    RCS-IUT (redundant copies)   0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
    EssOR-IUT                    0-link: N=-1.50 ∧ N=-0.50; multiradial: N=-0.50 ∧ N=-0.50 ⇒ no conclusion
```

The toolkit plugs in sample values and evaluates five readings that [EssLgc] §2.4 discusses: only the AND reading reaches a conclusion.

10.1 How the AND survives the algorithm

[EssLgc] §3.6–§3.10 then tracks the $\wedge$ through every step of Figure 6.1. Each step either keeps it (the descents (Stp2), (Stp3) replace data by equivalent algorithms on weaker data; (Stp5), (Stp6) do not change the possible value-group portions), or enlarges a set of possibilities, which only adds $\vee$'s *inside* one side and never across the Θ-link ($\dot{\vee}$, "(ExtInd1)"). The closed loop of (Stp7) brings both sides into one copy of $\mathbb{R}$, where the $\wedge$ becomes an inclusion of a number in a half-line. The danger [EssLgc] warns about most is (AOC3): wanting a proof that "taking log-volumes of pilot objects in the domain and codomain of the Θ-link yields the same real number". In the AND reading there is no such statement to prove; the two log-volumes live in different universes until the very last step.

OTAKU CORNER · 二股

In a harem romcom the protagonist is with heroine A *or* heroine B, and the whole season is about which. IUT is the rare series in which the protagonist is, for structural reasons, with A *and* B at once, because they live in different universes and never meet. Fans who insist "it has to be one of them" (the RCS reading) conclude the plot makes no sense. Fans who accept the AND find the ending forced: the only consistent epilogue is the one where the protagonist's height is bounded. Opinions on whether this ending was earned are in §12.

11 Analogy gallery たとえ話ギャラリー

Mochizuki explains the multiradial algorithm through at least six analogies of his own. None of them is a proof and each captures a different facet. Here they are side by side, followed by our less dignified additions.

11.1 The Gaussian integral [Alien §1, §3.8]

The classical computation of $I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$ has no elementary antiderivative to work with. The trick is to take *two identical but mutually alien copies* of the integral, one in x and one in y, multiply them, and pass to polar coordinates, where the radial and angular parts decouple:

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta = \pi, \qquad I = \sqrt{\pi}.$$

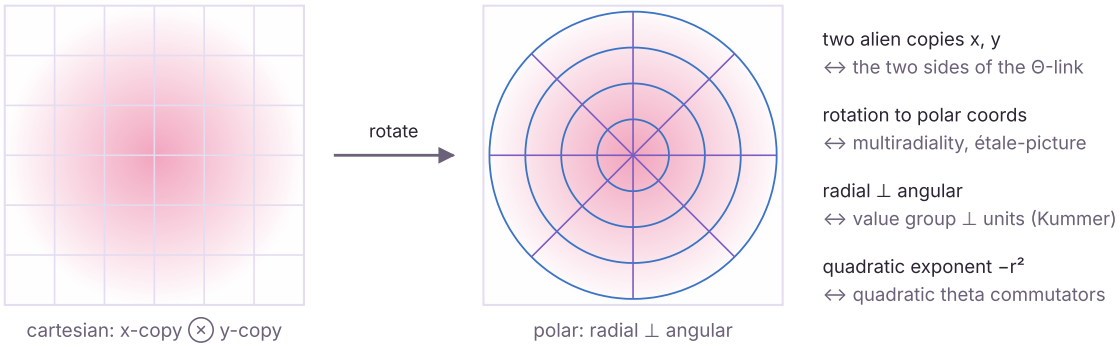


Figure 11.1. The Gaussian integral as a model of the multiradial algorithm ([Alien] Fig. 3.20.1). The cartesian grid is one copy of arithmetic per axis; the rotation that mixes them is only possible because the two copies are kept distinct ("alien") until then.

TOOLKIT · IUT RUN COMPUTEGAUSSIAN

```
ComputeGaussian  [Alien] §1 Steps 1-12 — the Gaussian integral and its IUT analogues
  in: —   out: √π
    Step  1 the naive approach e^{-x²} ↦ u fails: no elementary antiderivative      —   ↔ a single ring/scheme theory
cannot compute the height
    Step  2 partial integration does not help either                                —   ↔ no Frobenius morphism in char-
acteristic 0
    Step  3 introduce a SECOND copy of the integral: I² = ∫∫ e^{-x²-y²}  3.14159   ↔ two mutually alien copies of the
ring theory (the two sides of the θ-link)
    Step  4 regard them as coordinates of R²                                  3.14159   ↔ the log-theta-lattice as a
plane of copies
    Step  5 infinitesimals dx dy: the measure                                  3.14159   ↔ log-volumes
    Step  6 the Jacobian: how the measure transforms                           1.00000   ↔ compatibility of log-volumes
with the links
    Step  7 linear transformations: toral dilations, unipotent, rotations      1.00000   ↔ the dilation q ↦ q^{j²} of
the θ-link vs the rotation by the log-link
    Step  8 polar coordinates on R² \ (R≤0 × {0})                          1.00000   ↔ the coordinates (n, m) of the lat-
tice; the slit = the 'Kummer-detachment' indeterminacy
    Step  9 decoupling of the radial and angular parts                         1.00000   ↔ multiradiality: the radial
data is decoupled from the coric data
    Step 10 radial integral ∫_0^∞ e^{-r²} 2r dr = 1                        1.00000   ↔ the multiradial representation com-
putes the θ-pilot
    Step 11 angular integral ∫_0^{2π} (1/2) dθ = π                          3.14159   ↔ the holomorphic hull / the inde-
terminacies contribute the constant
    Step 12 √π: the result, with its 'error factor' from the slit           1.77245   ↔ −|log θ| ≥ −|log q| with the in-
determinacies as error term
    numerical ∫e^{-x²} = 1.772454, √π = 1.772454
```

11.2 Mochizuki's other analogies, in one table

analogy	what corresponds to what	source
Parallel transport / connections	coric data = base, radial data = fibres, multiradiality = a transport isomorphism lifting any path; (Ind1)–(Ind3) = monodromy; SHE = "parallel transport mechanism for the Θ-pilot".	[IUTchII] Rem 1.7.1; [IUTchIII] Rem 3.11.1 (iv)
Crystals	abstract F×μ-prime-strip vs its realizations in Hodge theaters ↔ a reduced char-p scheme vs its thickenings over $\mathbb{Z}_p$; working up to indeterminacies ↔ working modulo p.	[IUTchIII] Rem 3.11.1 (iii); [EssLgc] §3.10
Kodaira–Spencer map	rank-two bundle $\mathcal{E}$ as container for ω_E; transport across $S \times S$; the discrepancy bounds $\deg \omega_E$.	[Alien] §3.1 (v)

Faltings' isogeny invariance	the unusual case where a “global multiplicative subspace” exists and the comparison is exact: $\varepsilon = 0$.	[Alien] §2.3; [EssLgc] Ex 3.2.1; [Form26] §3
Frobenius mod p	a common ring structure compatible with the p-th power map: the thing IUT has to fake for $N = j^2$ over $\mathbb{Q}$.	[Form26] §3; [Alien] §2.4–§2.6
Fine vs coarse moduli	(Ind1)–(Ind3) describe a “formal quotient”; log-volumes are its “coarse space”, invariant under (Ind1), (Ind2).	[IUTchIII] Cor 3.12 proof, Step (x)
CPU and network cable	theta functions are hidden internal structures of the algorithm; the F×μ-prime-strip is the simple cable that connects two complicated computers.	[IUTchIII] Rem 3.11.1 (iv)
Analytic continuation	the multiradial representation is “analytic continuation” along the infinite H; the Θ-link gap is “miraculously rotated” onto the harmless log-link gap.	[Form26] §3; [EssLgc] §3.11
Upper half-plane	$GL_2(\mathbb{R})$-orbits of complex structures ↔ multiradial environments; classical Teichmüller dilation ↔ the Θ-link's dilation.	[Alien] Ex 3.2.1 (iii); [BogIUT]

11.3 Our additions: the multiradial algorithm as Japanese pop culture

IUT	pop culture	why it fits
Θ-link	*Your Name* body swap	shared body, unshared world; communication only by notes valid on both sides (§1)
Hodge theater	a parallel world line	a complete world, alien to the others; no direct contact
log-link	a daidōgei juggler	⊞ and ⊠ swap hands with every throw (§4)
Kummer theory	the cake and its photo	the photo travels, the cake does not (§4.3)
F×μ-prime-strip	an isekai protagonist's pockets	everything else stayed behind (§2)
mono-theta rigidity	a sync ratio that never drops	its cyclotome cannot be rescaled from outside (§6)
procession	a matsuri parade	each float adds one more label; reorderings are limited
(Ind1)	a gacha banner	288 outcomes for $l = 7$, same log-volume every pull
(Ind2)	swivelling Shinkansen seats	everyone rotates, nobody's weight changes (§7)
(Ind3)	the luggage rack	at least as big as the luggage, so the scale says “at most”
holomorphic hull	a furoshiki	one square cloth around odd shapes; its size is easy to state (§8)
log-volume	the sumo weigh-in	a public number everyone trusts (§9)
multiradial vs uniradial	karaoke in any key	the loudness survives every key change (§3)
AND vs OR	the two-heroine romcom	with both at once, in different universes (§10)
the 2018–2026 debate	a long-running series' finale discourse	everyone agrees on the episodes, not on whether the ending was earned (§12)

OTAKU CORNER · ネタバレ注意

Spoiler warning for anyone reading [IUTchIII] for the first time: the theta function, the hero of 180 pages, is deliberately forgotten in the last three ("hidden internal structures"). This is the IUT equivalent of a protagonist who saves the world and is then written out of the final episode so the side characters can sign the peace treaty. Mochizuki's justification is the same one a scriptwriter would give: the treaty only needs to know what was saved, not who saved it.

12 Where the debate stands (November 2026) 論争の現在地

NEUTRAL GROUND

This section reports what the participants have written, in their own words where possible, and maps each position onto the steps of this guide. It takes no side. The toolkit in this repository implements the *statements* of the theory as data types and bookkeeping; it cannot settle whether Corollary 3.12 is proved.

Almost all of the disagreement about IUT is concentrated on exactly the material of this guide: the passage from Theorem 3.11 to Corollary 3.12. Nobody seriously disputes the earlier anabelian results the theory consumes (Mochizuki's own 2026 report says the material of "Stages 3–5" has "never been a source of confusion/debate"). The timeline:

when	who	what	which step of this guide
Aug 2012	S. Mochizuki	[IUTchI–IV] posted.	all
Mar 2018	P. Scholze, J. Stix	After a week of discussions at RIMS, a report ("Why abc is still a conjecture") arguing that the abstract copies in the theory can be identified with one another without loss, that the Θ-link then forces an identification incompatible with $\underline{q} \mapsto \underline{q}^{j^2}$, and that the proof of Corollary 3.12 has a serious gap. Mochizuki published replies the same year.	§2.2, §8.2, §10
2021	PRIMS	[IUTchI–IV] published in a special issue of *Publ. RIMS* 57.	—
2022	S. Mochizuki	[EssLgc]: the AND/OR analysis of §10; the objection is attributed to a "redundant copies school" that identifies the two sides of the Θ-link.	§10
Mar 2025	K. Joshi	"Final Report on the Mochizuki–Scholze–Stix Controversy" (arXiv 2505.10568): argues the Scholze–Stix argument is mistaken but also that "Mochizuki's proof is also incomplete", since a key point is not clearly proved, "without clearly proving this, there is no way to provide a complete proof of [Theorem 3.11] (let alone [...] Corollary 3.12)"; maintains that abc follows from Joshi's own arithmetic Teichmüller theory.	§6
Apr 2026	S. Mochizuki et al.	[Form26]: progress report on a Lean formalization; Stage 1 is "[IUTchIII] Theorem 3.11 $\Longrightarrow$ Corollary 3.12", reorganized as 3.11 $\Longrightarrow$ 3.11.5 $\Longrightarrow$ 3.12, with skeletal Lean code for the last arrow.	§8.4
Jul 2026	Project LANA (J. Commelin, K. Kedlaya, Y. Hoshi, A. Topaz, F. Kato, leader)	Interim report on a Lean formalization (code released July 17, 2026). It gives no verdict, but says "At present, however, Project LANA has not yet reconstructed a proof of the required compatibility" and "there seems a point where a proof is not fully written", describing a "wall" about whether a certain diagram commutes, or a certain isomorphism is the identity, once (Ind1)–(Ind3) are allowed.	§7, §8.2 (xi-b)–(xi-d)

Jul 30, 2026	K. Joshi	“Comments on the LANA Project Report of Kato et al.”: critical of the report's methodology, arguing it rediscovers known problems, and again pointing to Joshi's own work.	—

12.1 The disagreement, mapped onto the algorithm

Put in the vocabulary of this guide, the positions differ on one question: **at step (xi), are the two routes of Figure 8.1 really computing the same thing, in the same copy of $\mathbb{R}$, with the blur on the correct side?**

- **Mochizuki's reading.** Yes, by construction: SHE and IPL say the output is expressible in column 1's ring structure and is linked to the fixed input; the closed loop of (Stp7) places both log-volumes in one container; everything else is an AND that was never broken. Asking for a separate proof that the two pilots' log-volumes agree is the error (AOC3).
- **The Scholze–Stix reading.** If the copies can be identified (they argue nothing is lost by doing so), the argument reduces to a comparison in which the indeterminacies must absorb the factor j^2, and then the inequality has no content. They locate the gap in the proof of Corollary 3.12.
- **Joshi's reading.** The copies cannot be naively identified (agreeing with Mochizuki against Scholze–Stix), but a needed property of the construction is not proved in [IUTchIII], so Theorem 3.11 itself is incomplete.
- **LANA's reading, as of July 2026.** No verdict; the formalization has reached a specific compatibility involving (Ind1)–(Ind3) that it has not yet been able to reconstruct from the papers.

What all four have in common is that the argument has been narrowed to a small part of the theory, the part this guide covers, and that formalization is now the route by which it will be tested. The toolkit's Lean port in `lean/` formalizes the data model and the arithmetic of the procession (§9.2), not the theorem.

OTAKU CORNER · 考察班

Every big series has a 考察班 (kōsatsu-han), the fan research squad that rewatches episodes frame by frame. IUT's squad is unusually distinguished, and for fourteen years it has been rewatching the same scene: the ten minutes between Theorem 3.11 and Corollary 3.12. The 2026 twist is that the squad now includes a proof assistant, which does not get tired, does not take sides, and does not accept “it is obvious from the definitions” as a frame. As of this writing, the season finale has been delayed.

13 Run it yourself 自分で動かす

Every toolkit box in this guide is live output, regenerated when the PDF is built. To reproduce it:

COMMANDS

```
make -C src                                          # build the C library, CLI and TUI
D=docs/initial-theta-data/frey-1-8-9.json            # or frey-1-2400-2401, frey-2-6436341-6436343, curve-11a1

src/iut pipeline --theta-json $D                     # stages 0–7; stage 5 is this guide
src/iut run RepresentMultiradially --theta-json $D   # Thm 3.11: (a)(b)(c), Ind1–Ind3
src/iut run EstimateLogVolume      --theta-json $D   # Cor 3.12 as numbered steps
src/iut run ChainAndSteps          --theta-json $D   # [EssLgc] §3.10 (Stp1)–(Stp8)
src/iut run StepEstimate           --theta-json $D   # [Alien] §3.7 (1est)–(12est)
src/iut run Fuzzify                --theta-json $D   # engulfing, (ExtInd1/2)
src/iut run ClassifyRadial                           # radial environments
src/iut tui --theta-json $D                          # the ncurses TUI: walk the lattice

(cd scripts/multiradial && npm ci) && node scripts/multiradial/build.js   # rebuild this PDF
```

TOOLKIT · IUT RUN STEPESTIMATE

```
StepEstimate  [Alien] §3.7 (1est)-(12est), (iv) — the estimate step by step; the leading term
  in: LogThetaLattice   out: numbers
    (1est) start with the q-pilot on the RHS (1,0): −|log q|                    -0.3139   [Alien] §3.7 (ii) (1est)
    (2est) the θ-link identifies it with the θ-pilot of the LHS (0,0)           -0.3139   (2est)
    (3est) the multiradial representation expresses the θ-pilot in RHS terms     15.5859   (3est); [IUTchIII] Thm
3.11
    (4est) Ind1-invariance: permutations of the procession                       288.0000   (4est)
    (5est) Ind2-invariance: Ism on the units                                       1.0000   (5est)
    (6est) Ind3 gives an UPPER bound on the θ-pilot's region                     15.5859   (6est)
    (7est) comparison only up to isomorphism class of O_k-modules                  0.0000   (7est)
    (8est) holomorphic hull = ⊗ of j+1 copies of log(O_k^×): log d^K terms      17.0507   (8est)
    (9est) log-volume = arithmetic degree                                        15.5859   (9est); [IUTchIII] Prop
3.9 (iii)
    (10est) log-Kummer on both sides                                               0.0000   (10est)
    (11est) the inequality −|log θ| ≥ −|log q|                                15.8998   (11est); [IUTchIII] Cor 3.12
    (12est) leading term (l²/48)(6 log d^K − log q), l ≈ √height              30.3259   (12est); [Alien] §3.7 (iv)
    leading term per j (j=1): 11.0533; sum over F_l^*: 46.7578 ≈ (l²/48)(6 log d^K − log q) = 30.3259; l ≈ √ht = 0.86
```

The web app (`scripts/serve.sh`, then open `htdocs/`) has a **Pipeline** view that runs the same C code compiled to WebAssembly and draws each stage, with an "exact Θ-data (JSON)" option that loads the four examples. The Lean port (`cd lean && lake build`) proves, among other things, that (Ind1) on packet j has order $(j+1)!$ (`card_Ind1`), the closed form of the procession-normalized upper end used in §9.2 (`thetaUpperBound_eq`), and that $C_\Theta \geq -1$ is equivalent to the bracket of Theorem 1.10 being non-negative (`CTheta_ge_neg_one_iff`).

to explore	where
the four examples, every stage	`docs/PIPELINE.md`, `docs/initial-theta-data/pipeline/*.txt`
all 73 processes	`src/iut procs`, `docs/PROCESSES.md`
all 448 identifiers	`docs/IDENTIFIERS.md`
the data model	`docs/DATA_MODEL.md`
4K infographics	`docs/*-infographic.png` (identifiers, mutations, prime-strips, Hodge theaters, initial Θ-data)
the light-hearted overview	`funfun_iut.pdf`
anabelian reconstruction, Kummer theory, cyclotomic rigidity	`anabelian_guide.pdf` (companion guide)
the papers	`pdf/`: [IUTchIII] pp. 153–186 for Theorem 3.11 and Corollary 3.12

14 Glossary and references 用語集と文献

term	meaning in one line
APT	algorithmic parallel transport: the transport is an algorithm runnable on both sides, not a map of regions
bi-coric	shared by both the vertical (log) and horizontal (Θ) arrows: G_v, $\mathcal{O}^{\times\mu}$, log-shells
BPS	basic prime-strip, [Form26]'s name for the F×μ-prime-strip
coric / radial	the shared underlying data / the richer structure over it
étale-like / Frobenius-like	rebuilt from Galois groups, portable / living in one Hodge theater, rigid
étale-picture	the D-Hodge theaters as spokes around the coric D⊢; its permutation symmetries give $(n, \circ) \cong (n', \circ)$

HIS	hidden internal structures: the theta functions, forgotten when applying Thm 3.11
holomorphic hull	smallest module over the ring of integers containing a region
(Ind1), (Ind2), (Ind3)	procession/Galois automorphisms; isometries of units; upper semi-compatibility
IPL	input prime-strip link: the output is linked to the (fixed) input prime-strip
Kummer-detachment	passing from Frobenius-like to étale-like data via Kummer isomorphisms
log-Kummer correspondence	the Kummer isomorphisms along one column, compatible with log-links up to (Ind3)
log-shell	$\frac{1}{2p}\log(\mathcal{O}_{K_v}^{\times})$: the additive container for logs of units
log-volume	procession-normalized volume of a region; = minus the arithmetic degree
multiradial	radial functor is full: any change of coric data lifts to the radial data
procession	$\{\mathcal{D}_0\} \to \{\mathcal{D}_0, \mathcal{D}_1\} \to \cdots$: labels added one at a time
q-pilot / Θ-pilot	$\underline{q}$ / $\{\underline{q}^{j^2}\}_{j=1..l^*}$ as objects of a global realified Frobenioid
SHE	simultaneous holomorphic expressibility: valid in the ring structures of both sides
tensor packet	tensor product of $j+1$ log-shells over $\mathbb{S}^{\pm}_{j+1}$

References

1. [IUTchI–IV] S. Mochizuki, *Inter-universal Teichmüller Theory I–IV*, Publ. RIMS 57 (2021). In `pdf/`. Theorem 3.11 and Corollary 3.12 are in [IUTchIII], pp. 153–186.
2. [Alien] S. Mochizuki, *Alien Copies, Gaussians, and Inter-universal Teichmüller Theory*. In `pdf/`; §1 (the Gaussian integral), §3.1–§3.2 (multiradiality), §3.7–§3.8 (log-volume estimates, dictionary).
3. [EssLgc] S. Mochizuki, *On the Essential Logical Structure of Inter-universal Teichmüller Theory in Terms of Logical AND "∧"/Logical OR "∨" Relations: Report on the Occasion of the Publication of the Four Main Papers on Inter-universal Teichmüller Theory* (2022; version of September 2023). In `pdf/`; Example 2.4.5, §3.10, §3.11.
4. [Form26] S. Mochizuki, *On the Formalization of IUT: a Preliminary Progress Report* (April 2026), joint work in progress with Y. Hoshi, G. Yamashita, Y. Yang et al. In `pdf/` and `leanform/`.
5. [EtTh] S. Mochizuki, *The Étale Theta Function and its Frobenioid-theoretic Manifestations*, Publ. RIMS 45 (2009).
6. [ExpEst] S. Mochizuki, I. Fesenko, Y. Hoshi, A. Minamide, W. Porowski, *Explicit estimates in inter-universal Teichmüller theory*, Kodai Math. J. 45 (2022). In `pdf/`.
7. [BogIUT] S. Mochizuki, *Bogomolov's proof of the geometric version of the Szpiro Conjecture from the point of view of inter-universal Teichmüller theory*. In `pdf/`.
8. [Pano] S. Mochizuki, *A Panoramic Overview of Inter-universal Teichmüller Theory*. In `pdf/`.
9. [SS18] P. Scholze, J. Stix, *Why abc is still a conjecture* (2018), and Mochizuki's 2018 reports in reply.
10. [Jo25] K. Joshi, *Final Report on the Mochizuki–Scholze–Stix Controversy*, arXiv:2505.10568 (2025), arxiv.org/abs/2505.10568.
11. [LANA26] Project LANA, *Interim Report on IUT Theory* (July 16, 2026), ncatlab.org/nlab/files/LANAProject-Report-July2026.pdf.
12. [Jo26] K. Joshi, *Comments on the LANA Project Report of Kato et al.* (July 30, 2026), linked from profiles.arizona.edu/person/kirti.
13. [Toolkit] This repository: `src/` (C library, CLI, TUI), `htdocs/` (web app), `lean/` (Lean 4 port), `docs/PIPELINE.md` (the four worked examples).

おつかれさまでした。 Thank you for riding the multiradial line to the end. Please take all your indeterminacies with you when leaving the train.